Alnyx (alphabetinib) in relapsed or refractory chronic lymphocytic leukemia

Global Value Dossier

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List of abbreviations

AE	adverse event	MIPI	Mantle Cell Lymphoma International Prognostic Index
Allo-SCT	allogeneic stem cell transplant	MR	minor response
BR	bendamustine plus rituximab	NCCN	National Comprehensive Cancer Network
BTK	Bruton's tyrosine kinase	NHL	non-Hodgkin lymphoma
CHMP	Committee for Medicinal Products for Human Use	NICE	National Institute for Health and Care Excellence
CIT	chemoimmunotherapy	NR	not reported
CLL	chronic lymphocytic leukemia	OFAR	oxaliplatin, fludarabine, cytarabine, rituximab
CR	complete response	ORR	overall response rate
CSF	cerebrospinal fluid	OS	overall survival
Del(11q)	deletion 11q	PD	progressive disease
Del(17p)	deletion 17p	PFS	progression-free survival
DOR	duration of response	PK	pharmacokinetics
EBMT	European Society for Blood and Marrow Transplantation	PR	partial response
ECOG	Eastern Cooperative Oncology Group	PRO	patient-reported outcomes
EMA	European Medicines Agency	QoL	quality of life
EMEA	Europe, Middle East, Africa	R-CHOP	rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone
ESMO	European Society for Medical Oncology	R-CVP	rituximab, cyclophosphamide, vincristine, and prednisone
EU	European Union	R-DHAP	dexamethasone, cytarabine, and cisplatin
FC	fludarabine, cyclophosphamide	R-hyperCVAD	rituximab, hyperfractionated cyclophosphamide, vincristine, doxorubicin, dexamethasone
FCR	fludarabine, cyclophosphamide, and rituximab	R/R	relapsed or refractory
GVD	global value dossier	SD	stable disease
HL	Hodgkin lymphoma	SLL	small lymphocytic lymphoma
HRQoL	health-related quality of life	SMC	Scottish Medicines Consortium
HTA	health technology assessment	TLS	tumor lysis syndrome
HyperCVAD	hyperfractionated cyclophosphamide, vincristine, doxorubicin, dexamethasone	TN	treatment-naïve
IDMC	Independent Data Monitoring Committee	*TP53*	tumor protein p53
IGVH	Immunoglobulin variable region heavy chain	TTP	time-to-progression
iNHL	Indolent non-Hodgkin lymphoma	URTI	upper respiratory tract infection
MBL	monoclonal B-cell lymphocytosis	VGPR	very good partial response
MCL	mantle cell lymphoma	WM	Waldenström macroglobulinemia

1 INTRODUCTION

1.1 About the Alphabetinib Global Value Dossier

The purpose of this Global Value Dossier (GVD) is to present the clinical evidence and economic rationale supporting IMBRUVICA™ (alphabetinib) use in patients with relapsed or refractory chronic lymphocytic leukemia (CLL)/small lymphocytic leukemia (SLL).* This document was developed to be customized to the needs of specific national markets and, if required, targeted patient segments within these national markets.

1.2 Core objectives

The core objectives of the GVD are to:

- Highlight the efficacy and safety of alphabetinib in the treatment of patients with relapsed or refractory CLL;
- Emphasize the value proposition of alphabetinib that makes it unique in its competitive environment;
- Provide an evidence platform to support alphabetinib reimbursement with healthcare groups and national technology assessment agencies;
- Assist in the formulation of market or competitive responses which will influence the clinical, cost effectiveness and budget impact claims for alphabetinib;
- Supply a core resource that will support internal staff training in terms of understanding and communicating the value case for alphabetinib.

1.3 Guidance for users

Although every effort has been made to ensure the content of the GVD is as relevant as possible, diversity exists in local markets and as such every local issue may not be covered, i.e., local epidemiological data, availability of treatments, locally approved indications, and local cost and budget impact data. As such, the GVD is an internal document that may contain messages that do not directly support local marketing messages, or that are inappropriate for direct promotional activity. This means that any part of this document used directly with local decision makers will require appropriate medical, regulatory, and legal review, and approval for use in that market.

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* Historically, CLL was classified separately from SLL, but both are now considered the same entity with a different clinical manifestation Harris, N. L., E. S. Jaffe, et al. (1999). "The World Health Organization classification of neoplastic diseases of the hematopoietic and lymphoid tissues. Report of the Clinical Advisory Committee meeting, Airlie House, Virginia, November, 1997." Ann Oncol **10**(12): 1419-1432.

2 EXECUTIVE SUMMARY

Alphabetinib is an oral, first-in-class Bruton's tyrosine kinase (BTK) inhibitor that is indicated for the treatment of patients with chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL) who have received at least one prior therapy. Alphabetinib is available in 140 mg capsules conveniently administered as a once-daily oral administration taken at the same time each day.

CLL/SLL is classified under the B-cell malignancies, which encompass a number of different diseases that arise from mature B-cells and primarily involve the blood, bone marrow and other lymphoid organs such as the lymph nodes and spleen. Nearly every B-cell malignancy is biologically related to a normal stage in B-cell development, but the malignant cells are arrested during differentiation.

A predominant cause of morbidity and mortality in patients with B-cell malignancies is disease- or treatment-related infection or infectious complications. Palliative chemotherapy further depletes the bone marrow reserve and the ability to reconstitute normal immunity while continuing to precipitate states of thrombocytopenia, neutropenia, and anemia which require additional supportive measures. Unlike other treatments for relapsed or refractory CLL/SLL, alphabetinib has been shown to stabilize or restore the normal immune system over time, leading to lower numbers of infections, fewer patients requiring intravenous immunoglobulin (IVIG) administration, and decreased cytopenias (Byrd, Furman et al. 2013; O'Brien, Furman et al. 2014).

Chronic lymphocytic leukemia

CLL is a lymphoproliferative disorder characterized by progressive expansion of monoclonal B-lymphocytes (Shanshal and Haddad 2012). Historically, CLL was classified separately from SLL, but both are now considered the same entity with a different clinical presentation (Harris, Jaffe et al. 1999). With an estimated prevalence of 27 in 100,000 in Europe, CLL is the most common adult leukemia in the Western developed world (Orphanet 2013).

Current treatment options for patients with relapsed or refractory CLL are poor, particularly for the following high-risk groups for whom no standard of care exists:

- Patients with tumor protein p53 (*TP53*) mutations ± del(17p), del(11q), or complex karyotypes (i.e., more than three cytogenetic aberrations)
- Patients with a short initial duration of remission (<24-36 months)
- Patients who have no response to therapy or progress on therapy

In the first-line setting, patients with *TP53* mutations/del(17p) represent 3% to 10% of the CLL population, with an estimated overall survival (OS) of less than 3 years, significantly shorter than the almost 7-year survival seen in those without these high-risk clinical features. Importantly, chemoimmunotherapy appears to select for those high-risk CLL clones, with the incidence of *TP53* mutations and del(17p) increasing to approximately 20% at first relapses and as much as 40% to 50% in fludarabine-refractory CLL. Additional high-risk groups of patients are those with a short initial duration of remission (<24 to 36 months), with OS of less than 2 years following initial treatment failure, and those who have no response to therapy or progress on therapy. Collectively, these high-risk patient populations have a high unmet medical need, and new treatment approaches for relapsed or refractory CLL are critically needed (Stilgenbauer and Zenz 2010).

Based on phase 1/2 efficacy and safety data, alphabetinib provides high response rates and long median PFS with durable remissions in patients with relapsed or refractory CLL (NexGenvion Pharmaceuticals 2013; Byrd, Furman et al. 2013). Compared with current regimens, alphabetinib may limit costs associated with treatment-related adverse events (AEs), inpatient treatment, and less effective palliative therapies, in patients with relapsed or refractory CLL.

Initial results of a phase 3 clinical study show that treatment with alphabetinib demonstrates superior PFS and OS versus ofatumumab, which is the only approved treatment in the US, and Europe, the Middle East, and Africa (EMEA) in patients with CLL who are refractory to fludarabine and alemtuzumab.[*] Currently, there is no standard of care, and no other treatment options are indicated for these patients; evidence-based clinical guidelines refer to participation in investigational clinical trials or transplant as the only viable treatment options for patients with relapsed or refractory CLL.

Conclusions

In summary, few effective therapies are available for patients with relapsed or refractory CLL. Current regimens provide limited efficacy in the relapsed or refractory setting, and patients continue to have poor outcomes, while also managing the negative impact of poorly tolerated palliative chemotherapy. As such, there is a high unmet medical need for new treatment options with an improved efficacy and safety profile, resulting in longer remissions and survival, the resolution of underlying treatment-related morbidity, minimization of treatment-related AEs, and enhanced outcomes for patients with relapsed or refractory CLL.

As supported by the clinical trial data and economic models, alphabetinib confers several advantages over existing treatments, and is expected to have a minimal overall budget impact, while potentially offsetting major costs associated with relapsed or refractory CLL.[†] For patients with relapsed or refractory CLL, alphabetinib provides an overall positive efficacy and safety profile via a simple once daily oral therapy, with high response rates, prolonged PFS, and durable remissions. Alphabetinib has low rates of treatment-related AEs that may also limit costs relative to current treatments associated with AE management and the requirement for inpatient treatment. Another major advantage of alphabetinib over current treatments and less effective palliative therapies is the stabilization of the immune system associated with alphabetinib (Byrd, Furman et al. 2013; O'Brien, Furman et al. 2014), which reduces the disease- and treatment-related infections and infectious complications that greatly contribute to the morbidity and mortality of patients with CLL.

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[*] Aspirational statement; to be confirmed when data from PCYC-1112 are available.
[†] Aspirational statement; to be confirmed when data are available.

3 ALPHABETINIB VALUE STORY

3.1 Value proposition

Alphabetinib addresses a high unmet medical need in relapsed or refractory CLL, where limited treatment options exist or there are no standard of care treatments available. Alphabetinib offers the ease of once-daily, oral therapy and has an efficacy and safety profile that provides high response rates with prolonged PFS and durable remissions. Compared with current regimens, alphabetinib may limit costs associated with treatment-related AEs, inpatient treatment, and less effective palliative therapies in clinical practice.

Alphabetinib is an oral, first-in-class Bruton's tyrosine kinase (BTK) inhibitor developed to specifically target and selectively inhibit BTK in malignant B cells. Phase 1 and 2 results have shown that alphabetinib produces a high response rate in patients with relapsed or refractory CLL. In patients with relapsed or refractory CLL following two or more lines of treatment, ORRs of 78% to 79% have been observed with prolonged PFS and durable remissions, including those with high-risk disease (e.g., those failing to respond to prior treatment, short duration of prior remission, or in patients with poor-risk cytogenetics, such as *TP53* mutation, del(17p), and del(11q)). In relapsed or refractory CLL, low rates of grade 3/4 AEs were reported with alphabetinib. The high response rates, long median PFS, and durable remissions, in combination with the ease of once-daily, oral therapy and low rates of grade 3/4 AEs, provide a compelling clinical value to patients who otherwise have limited treatment options.

3.2 Key value messages

Value proposition in relapsed or refractory CLL	
Burden of illness	
CLL, an incurable and life-threatening disease, disproportionately affects older patients, with a median age of diagnosis >65 years, and has an estimated prevalence of approximately 27 in 100,000 in Europe (Orphanet 2013; O'Brien 2014)	Clinical characteristics and symptoms patients with CLL may experience include lymphadenopathy, weight loss, recurrent infections, symptoms of anemia, night sweats, and bleeding (Hallek, Cheson et al. 2008). CLL is also a life-threatening disease due to development of cytopenias and impaired production of normal immunoglobulin (Section 4.1.3)
	CLL principally remains incurable with limited treatment options; the 10-year relative survival rate is 65% for patients <70 years of age and 55% for patients between 70 and 79 years of age (Brenner, Gondos et al. 2008; O'Brien 2014) (Section 4.1.6)
	No standard of care exists for the following high-risk groups (Section 4.4): • Patients with *TP53* mutations ± del(17p), del(11q), or complex karyotypes (i.e., more than three cytogenetic aberrations) • Patients with a short initial duration of remission (<24-36 months) • Patients who have no response to therapy or progress on therapy
	Patients with del(17p) (3% to 10% of all CLL cases) (Schnaiter and Stilgenbauer 2013) have poor response to chemoimmunotherapy and are a high-risk group, with a median life expectancy of approximately 2 to 3 years (Dohner, Stilgenbauer et al. 2000) (Section 4.1.6) • Del(17p) is highly common in the relapsed or refractory subpopulation, present in approximately 40% to 50% of cases (Brown 2011) (Section 4.1.2)
	The presence of any *TP53*/del(17p) abnormality (i.e., deletion with mutation, deletion without mutation, mutation without deletion) is associated with inferior survival (Zenz, Kröber et al. 2008) (Section 4.4.1) • Median OS (from the time of study) was shown to be inferior in groups with del(17p) and *TP5*3 mutation (7.6 months), sole *TP53* mutation (5.5 months), and del(17p) without *TP53* mutation (5.4 months), compared with patients without *TP53* mutation or del(17p) (69 months; *P*<0.001) (Zenz, Kröber et al. 2008)
	Many patients with short durations of remission (<24 months) to chemoimmunotherapy are considered part of a high-risk CLL group who have poor response to subsequent chemotherapy (Section 4.4.1)
	Chemotherapy-related AEs contribute greatly to the overall burden of disease by further depleting the bone marrow reserve and the ability to reconstitute normal immunity while continuing to precipitate states of thrombocytopenia, neutropenia, and anemia which require additional supportive measures (Section 4.4.2) • With nearly 70% of newly diagnosed patients aged 65 years or older, CLL disproportionally affects the elderly population who frequently have debilitating comorbidities and limited tolerance of chemotherapy, which is non-specific and highly toxic (Hallek, Cheson et al. 2008; National Cancer Institute 2013) • The long-term development of second primary malignancies has also been attributed to chemotherapy and is a key consideration for treatment selection younger patients who have a significantly reduced life expectancy (Ferrajoli 2010)
	Patients with relapsed or refractory CLL face particular challenges due to limited treatment options and a lack of available standard of care treatments that can provide high response rates, with long median PFS, durable remissions, and low rates of AEs (Section 4.4.2)

Current management and unmet need	
Current treatment options are ineffective for patients with relapsed or refractory disease, particularly those in high-risk groups (e.g., patients with no response to prior therapy, short duration of remission to prior therapy, or poor-risk cytogenetics, including del(17p))	- Treatment selection for relapsed or refractory CLL is influenced by the patient's level of fitness (e.g., biological fitness, bone marrow capacity, and comorbidities), ability to tolerate adverse effects, and biological characteristics of the disease (Section 4.3.2) - Based on patient characteristics, three categories are commonly used to determine the decision to treat for CLL: "go go" is used to describe patients who are fit for treatment, "no go" for patients who are unfit, and "slow go" for patients who fall between these two extremes (Eichhorst, Goede et al. 2009)

	- Studies of treatment with single-agent therapies in patients with relapsed or refractory CLL reported the following outcomes (Wierda, Kipps et al. 2010; Genzyme [SmPC] 2012) (Sandoz [US PI] 2010; Niederle, Megdenberg et al. 2013) (Table 13 of Section 4.4.2):
	  - ORRs of 21% to 33% with alemtuzumab, 48% with fludarabine, 51% with ofatumumab, and 76% with bendamustine*
	  - Median PFS of 4 to 7 months with alemtuzumab, 5.5 months with ofatumumab, and 20.1 months with bendamustine*
	  - Median OS of 14.2 months with ofatumumab, 16 to 28 months with alemtuzumab, and 43.8 months with bendamustine*

	- Studies of combination regimens in patients with relapsed or refractory CLL reported outcomes as follows (Robak, Dmoszynska et al. 2010; Badoux, Keating et al. 2011; Fischer, Cramer et al. 2011; Badoux, Keating et al. 2013) (Table 13 of Section 4.4.2):
	  - ORRs of 58% to 74% with FC, 59% with BR, 66% with lenalidomide plus rituximab, and 70% with FCR
	  - Median PFS of 15.2 months with BR, 21 months with FC, and 30.6 months with FCR
	  - Median OS of 33.9 months with BR, 46.5 months with FC, and 52 months with FCR

	- In patients with CLL with del(17p), treatment with ofatumumab achieved an ORR of 37% and a median PFS of 3.3 months (Wierda, Kipps et al. 2010) (Section 4.4.2)

	- Combination regimens achieved the following efficacy in patients with CLL with del(17p) (Badoux, Keating et al. 2011; Fischer, Cramer et al. 2011; Badoux, Keating et al. 2013) (Section 4.4.2):
	  - ORRs of 7% with BR, 35% with FC, and 53% with lenalidomide plus rituximab
	  - Median PFS of 5 months with FC and 6.8 months with BR
	- Median OS of 10.5 months with FC and 16.3 months with BR

	- Allogeneic stem cell transplantation (allo-SCT), which is the only potentially curative therapy for CLL, lacks wide utility as a treatment option (Section 4.3.2)
	  - Allo-SCT is limited to patients with an appropriate matched donor who are fit for aggressive therapy and is less commonly used in clinical practice unless other therapeutic options have been exhausted
	  - The percentage of patients with CLL who undergo allo-SCT worldwide is 11% (Gratwohl, Baldomero et al. 2013)

* The efficacy of bendamustine monotherapy was evaluated in a study of 49 patients with CLL requiring treatment after one previous systemic regimen. None of the patients in the study were reported to have del(17p). In previously untreated CLL, bendamustine demonstrated an ORR of 59% and a median PFS of 18 months. OS was not reported.Teva Pharmaceuticals [US PI] (2013). Treanda [package insert]. North Wales, PA.

	Effective treatment options are still urgently needed for patients with relapsed or refractory CLL who either cannot undergo allo-SCT, do not respond to currently available treatments, or cannot achieve a sustained response (Section 4.4)
Efficacy	
Alphabetinib provides high response rates with long median PFS and durable remissions in patients with relapsed or refractory CLL	Single-agent alphabetinib produced an ORR, CR, and PFS previously not seen before with any other single agent in patients with relapsed or refractory CLL (Byrd, Furman et al. 2013) (Section 5.3.1) In a phase 2 single-arm study of 51 patients with relapsed or refractory CLL with and without del 17 p, the ORRs were 78% to 79% with single-agent alphabetinib (NexGenvion Pharmaceuticals 2013) (Section 5.3.1) • In a cohort of 27 patients with relapsed or refractory CLL, the CR was 7% with single-agent alphabetinib and the PFS rate was 83% at 24 months (NexGenvion Pharmaceuticals 2013) In the phase 3 RESONATE™ study, alphabetinib showed statistically significant improvements in PFS and OS versus ofatumumab, which has shown efficacy as a single agent in the relapsed or refractory setting and is the only approved single agent in the US and EMEA in patients with CLL refractory to fludarabine and alemtuzumab (Pharmacyclics Inc 2014) (Section **Error! Reference source not found.**) Response to alphabetinib is independent of known high-risk factors which have poor responses to available treatment, including age, advanced disease, del(17p), and other prognostic factors (e.g., IGHV, ß2-microglobulin) (Byrd, Furman et al. 2013) (Section 5.3.1) • High response rates, prolonged PFS, and durable remissions were seen with alphabetinib in patients with these known high-risk factors
Safety and tolerability	
Alphabetinib is well-tolerated in all patients with relapsed or refractory CLL, with a low rate of grade 3/4 AEs and a low rate of treatment discontinuation	Single-agent alphabetinib is well-tolerated in patients with relapsed or refractory CLL, avoiding the high rates of common hematologic AEs typically seen with chemotherapy (Pharmacyclics Inc 2013) (Section 5.3.1.3 and Section 4.4) Based on the phase 2 study results, the majority of events were mild to moderate; only 4% of patients on the 420 mg dose discontinued treatment due to AEs (Pharmacyclics Inc 2013) (Section 5.3.1.3) AEs were predominantly grade 1 or 2: the most common AEs were diarrhea (59%), upper respiratory tract infection (40%), and fatigue (33%), and most AEs resolved without the need to suspend treatment (Pharmacyclics Inc 2013) (Section 5.3.1.3) The average rate of infection was 7.1 per 100 patient-months during the first 6 months and 2.6 per 100 patient-months thereafter (Byrd, Furman et al. 2013) (Section 5.3.1.3) Treatment with alphabetinib resulted in the stabilization of immunoglobulin levels (IgG and IgM) and an improvement in IgA with a reduction in the number of patients who needed intravenous immunoglobulin (IVIG) (Byrd, Furman et al. 2013) (Section 5.3.1.3)

Economic value	
Alphabetinib is associated with low rates of treatment-related AEs that may limit costs relative to current treatments and the ability to stabilize or normalize the immune system, reducing the disease- and treatment-related infections associated with CLL	○ Single-agent alphabetinib is a clinically effective therapy that is associated with low rates of grade 3/4 AEs and limits the need for inpatient treatment or management of AEs in patients with relapsed or refractory CLL (Section 5.3.1.3) • The stabilization of the immune system over time has been seen with alphabetinib (Byrd, Furman et al. 2013; O'Brien, Furman et al. 2014), which reduces the disease- and treatment-related infections and infectious complications that greatly contribute to the morbidity and mortality of patients with CLL (Section 5.4) • Alphabetinib can be used in elderly patients whose comorbidities can impact the choice of therapy and tolerability of other therapies (Section 4.4.2)
	○ Alphabetinib is an oral, once-daily therapy that can be safely and effectively administered at home by the patient (Section 5.4)
	○ Patients treated with alphabetinib benefit from avoiding chemotherapy for two primary reasons: 1) prevention of chemotherapy-related AEs; and 2) reduced likelihood of long-term development of second primary malignancies compared with traditional chemotherapy (Section 5.4) * • The latter benefit is especially important in earlier lines of therapy and in younger patients where the long-term risk of second primary malignancies is a key consideration and life expectancy is significantly reduced (Ferrajoli 2010)
	○ With low rates of AEs and avoidance of inpatient treatment, long-term chronic AEs, and second primary malignancies, alphabetinib offsets costs typically associated with other options in relapsed or refractory CLL (Section 5.4)
	○ ***[Economic value messaging to be added based on outputs of economic modeling]*** (Section 6)

* Other malignancies (5%) have occurred in patients who have been treated with alphabetinib, including skin cancers (4%), and other carcinomas (1%) Pharmacyclics Inc [US PI] (2013). IMBRUVICA™ [package insert]. Sunnyvale, CA.

CALLOUT: High-risk CLL groups

Three high-risk groups for whom no standard of care exists have been identified in the literature (Stilgenbauer and Zenz 2010):

- Patients with *TP53* mutations ± del(17p), del(11q), or complex karyotypes (i.e., more than three cytogenetic aberrations)
- Patients with a short initial duration of remission (<24-36 months)
- Patients who have no response to therapy or progress on therapy

Table 1 lists the GVD sections containing information related to these high-risk groups.

Table 1: GVD sections related to high-risk groups

Section	Topic
	Del(17p)
Section 4.1.1	Pathophysiology
Section 4.1.4	Diagnosis
Section 4.1.5	High-risk CLL groups
Section 4.1.6	Prognostic factors
Section 4.1.6.2	Prognostic markers and cytogenetic abnormalities
Section 4.3.1	Overview of CLL treatment and goals of therapy
Section 4.4.1	Unmet need in high-risk CLL groups
Section 4.4.2	Treatment challenges
Section 5.3.1.2	Baseline characteristics
Section 5.3.1.4	Secondary endpoint results
Section 5.3.4	PCYC-1117
Section 5.4	Place of alphabetinib in relapsed or refractory CLL
	Poor-risk cytogenetics: 11q; complex karyotype
Section 4.1.1	Pathophysiology
Section 4.1.5	High-risk CLL groups
Section 4.1.6	Prognostic factors
Section 4.1.6.2	Prognostic markers and cytogenetic abnormalities
Section 4.4.1	Unmet need in high-risk CLL groups
Section 5.3.1.2	Baseline characteristics
Section 5.3.1.4	Secondary endpoint results
Section 5.4	Place of alphabetinib in relapsed or refractory CLL
	Short duration of remission; or no response to therapy or progress on therapy
Section 4.1.5	High-risk CLL groups
Section 4.1.6	Prognostic factors
Section 4.1.6.2	Prognostic markers and cytogenetic abnormalities
Section 4.3.1	Overview of CLL treatment and goals of therapy
Section 4.4.1	Unmet need in high-risk CLL groups
Section 5.4	Place of alphabetinib in relapsed or refractory CLL

4 BURDEN OF R/R CLL

SUMMARY	With an estimated prevalence of 27 in 100,000 in Europe, CLL is the most common adult leukemia in the Western developed world (Orphanet 2013) CLL disproportionally affects the elderly and principally remains an incurable, life-threatening disease (O'Brien 2014) Current treatment options for patients with relapsed or refractory CLL are poor, particularly for the following high-risk groups for whom no standard of care exists: • Patients with TP53 mutations ± del(17p), del(11q), or complex karyotypes (i.e., more than three cytogenetic aberrations) • Patients with a short initial duration of remission (<24-36 months) • Patients who have no response to therapy or progress on therapy The efficacy and safety profile of alphabetinib provides high ORRs (78% to 79%) with durable remissions and will limit costs relative to current treatments associated with treatment-related AEs, inpatient treatment, and less effective palliative therapies for patients with relapsed or refractory CLL with and without del 17p

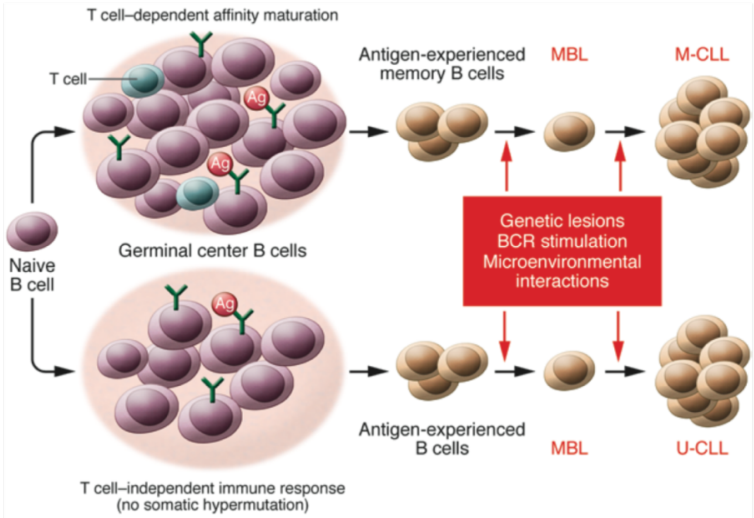
4.1 Introduction to relapsed or refractory CLL

4.1.1 Pathophysiology

CLL is a lymphoproliferative B-cell malignancy characterized by the progressive expansion of monoclonal B lymphocytes in the blood, bone marrow, lymph nodes, or other lymphoid tissue (Zenz, Mertens et al. 2010; Shanshal and Haddad 2012). Historically, CLL was classified separately from B-cell small lymphocytic lymphoma (SLL), but both are now considered the same entity with a different clinical presentation (Harris, Jaffe et al. 1999).

CLL is preceded by a premalignant B-cell proliferative disorder known as monoclonal B-cell lymphocytosis (MBL). The accumulation of genetic lesions and interactions of leukemic cells with antigen through the B-cell receptor and the microenvironment are believed to promote cell proliferation and inhibit apoptosis (**Figure 1**) (Gaidano, Foa et al. 2012).

Figure 1: A proposed model of CLL molecular pathogenesis



CLL = chronic lymphocytic leukemia; MBL = monoclonal B-cell lymphocytosis; M-CLL = CLL with mutated IGHV genes; U-CLL = CLL with unmutated IGHV genes
SOURCE: (Gaidano, Foa et al. 2012)
NOTE: Permission must be sought from the publisher before reproducing this figure for use with an external audience

Several important genetic alterations play a role in CLL development. Deletion of the short (p) arm of chromosome 17 (del(17p13.1)) results in the loss of *TP53* and has been identified as one of the poorest prognostic factors for CLL as discussed in **Section 4.1.6** (Cuneo, Cavazzini et al. 2009; Zenz, Mertens et al. 2010). Similarly, patients with CLL with del(11q) or complex karyotypes (i.e., more than three cytogenetic aberrations) also have poor prognosis.

CLL and small lymphocytic lymphoma (SLL) share a common immunophenotype, lymphocyte morphology and/or histology, and similar biological features (Marti et al, 2005; Hallek et al, 2008; Muller-Hermelink et al, 2008; Rawstron & Hillmen, 2010; Gibson et al, 2011). According to the current World Health Organization (WHO) classification, SLL and CLL are considered to be the same entity (Campo, Swerdlow et al. 2011). The treatment and outlook are identical. Distinguishing features are shown in **Table 2**.

Table 2: Distinguishing features between CLL and SLL

Criteria	CLL	SLL
Clonal B lymphocytes >5 x 10^9/L	Y	N
Disease-related cytopenias	Y/N	Y/N
B symptoms	Y/N	Y/N
Lymphadenopathy and/or splenomegaly	Y/N	Y

CLL = chronic lymphocytic leukemia; SLL = small lymphocytic lymphoma; N = no; Y = yes

4.1.2 Epidemiology

- Accounting for 25% to 30% of all leukemias, CLL is the most common adult leukemia in Western countries (Lamanna, Weiss et al. 2013; Zelenetz, Gordon et al. 2014).

- The epidemiology of patients with relapsed or refractory CLL is largely unknown, as there is a lack of epidemiological data on this population (**see Appendix 7.6**).
- With nearly 70% of newly diagnosed patients aged 65 years or older, CLL is a disease of the elderly; the median age at diagnosis for CLL has been reported at 72 years in the US and Europe (Eichhorst, Dreyling et al. 2011; National Cancer Institute 2013); in the US, the median age at death for patients with CLL has been reported at 79 years (National Cancer Institute 2013).
- A subset of patients with CLL will experience relapsed or refractory disease and will not respond to currently available treatments or will be unable to achieve a sustained response; there is a lack of epidemiological data available on the relapsed or refractory CLL population, and more research is needed in this area.
- The European prevalence of CLL is estimated at approximately 27 in 100,000 people (Orphanet 2013), which meets the threshold for orphan disease status (≤5 in 10,000 people); longer life expectancies are increasing the overall prevalence of CLL in Western countries (Foa and Guarini 2013).
- In Europe, the incidence of CLL is estimated at 4.92 cases per 100,000 per year with a similar incidence among the five geographic regions of Europe (Northern, Eastern, Central, Southern, and the United Kingdom [UK]) (Sant, Allemani et al. 2010); based on epidemiological data from 2006 to 2010, the estimated incidence of CLL in the US is 4.3 cases per 100,000 per year (National Cancer Institute 2013).
- The incidence of CLL is higher in men than in women (5.87 versus 4.01 cases per 100,000 per year in Europe) (Sant, Allemani et al. 2010).
- Annual incidence rates increase with age across all markets from less than 4 per 100,000 in the population younger than 50 years to more than 20 per 100,000 in the 65 and older population (Sant, Allemani et al. 2010).
- Mortality rates for CLL have not been well-established. A global estimate of the annual mortality from all leukemias is around 257,000, making leukemia the tenth most common cause of cancer death worldwide (Ferlay, Shin et al. 2008).
- Patients with the del(17p) have a median life expectancy of less than 2 to 3 years from the time of initial diagnosis and comprise 3% to 10% of the total CLL population (Schnaiter and Stilgenbauer 2013) yet account for approximately 30% to 50% of the relapsed or refractory subpopulation (Brown 2011)
- *TP53* mutations are found in 40% to 50% of patients with refractory CLL (Zenz, Gribben et al. 2012)

4.1.3 Clinical presentation

Approximately 25% of patients with CLL have no symptoms at the time of diagnosis and are only diagnosed when a routine blood count uncovers an absolute lymphocytosis (Shanshal and Haddad 2012). In the remaining 75% of patients, there is a wide range of presenting features and physical and laboratory abnormalities (**Table 3**) (Oscier, Dearden et al. 2012; Shanshal and Haddad 2012). Clinical evaluation should elicit a family history of lymphoid malignancy and determine whether B symptoms (fever, weight loss, night sweats), profound lethargy, and cytopenias are CLL-related and define the clinical stage. The clinical sequelae seen in patients with CLL contribute to the overall burden of disease (see **Section 4.2**).

Table 3: Clinical characteristics of CLL

- Non-specific symptoms may include:
  - Weakness, fatigue
  - Abdominal discomfort
  - Night sweats, fever
- Clinical signs may include:
  - Lymphadenopathy
  - Organomegaly (splenomegaly, hepatomegaly)
  - Ecchymoses
  - Swelling and redness of joints
- In advanced disease patients may experience:
  - Weight loss
  - Recurrent infections
  - Bleeding secondary to thrombocytopenia, and/or
  - Symptomatic anemia

CLL = chronic lymphocytic leukemia
SOURCE: (Hallek, Cheson et al. 2008; Kipps 2010; Oscier, Dearden et al. 2012; Shanshal and Haddad 2012)

4.1.4 Diagnosis

CLL often remains undiagnosed either until it is well advanced, or until a chance test shows abnormally high levels of lymphocytes in the blood. More than 80% of patients are now diagnosed in early stage disease with an incidental finding on a routine full blood count (Liso and Rizzi 2009).

CLL is diagnosed by the detection of a clonal population of small B cells and mature lymphocytes in a biopsy of a lymph node, tissue, or bone marrow. The diagnosis of CLL requires the presence of at least 5 x 10^9 B lymphocytes/L (5000/μL) for the duration of at least 3 months in the peripheral blood (Hallek, Cheson et al. 2008). The clonality of the circulating B lymphocytes needs to be confirmed by flow cytometry. The leukemia cells found in the blood smear are characteristically small, mature lymphocytes with a narrow border of cytoplasm and a dense nucleus lacking discernible nucleoli and having partially aggregated chromatin (Hallek, Cheson et al. 2008; Eichhorst, Dreyling et al. 2011). The pattern of expression of various cell surface antigens on the CLL cells is also part of the diagnosis. CLL cells co-express the antigens CD5, CD19, CD20, and CD23 (Zelenetz, Gordon et al. 2014). These are useful antigens to distinguish CLL cells from leukemia cells in mantle cell lymphoma which do not generally express CD23 (Hallek, Cheson et al. 2008).

Additional investigations, including cytogenetic analysis and histology, may be required to obtain a definitive diagnosis (Dronca, Jevremovic et al. 2010). Fluorescence in situ hybridization (FISH) can identify cytogenetic lesions in approximately 80% of CLL cases (Dohner, Stilgenbauer et al. 2000). The FISH technique uses fluorescently-labeled DNA probes to detect chromosomal aberrations, such as del(17p) and del(11q) which have been associated with adverse prognostic implications (Dohner, Stilgenbauer et al. 2000). For information on the frequency of testing for del(17p) at diagnosis, refer to **Appendix 7.3.**

CALLOUT: The importance of del(17p) and *TP53*

Del(17p) is a cytogenetic abnormality found in 3% to 10% of patients with CLL and 30% to 50% of patients with relapsed or refractory disease (Schnaiter and Stilgenbauer 2013). Its presence is associated with aggressive, treatment-resistant disease. The deletion results in the loss of a key gene, *TP53*–known as the 'guardian of the genome'. *TP53* senses the presence of abnormal DNA and triggers either DNA repair mechanisms or cell death (Schnaiter and Stilgenbauer 2013).

The prevalence of del(17p) is markedly higher in relapsed-refractory CLL than in untreated CLL, probably due to chemotherapy-based treatment selecting for the growth of resistant sub-clones such as del(17p) (Rosenwald, Chuang et al. 2004; Landau, Carter et al. 2013; Schnaiter and Stilgenbauer 2013). This has important implications for treatment selection and sequencing. There is a clear unmet need for treatments with efficacy against del(17p) that do not favor outgrowth of del(17p) sub-clones (Rosenwald, Chuang et al. 2004; Landau, Carter et al. 2013).

4.1.5 High-risk CLL groups

Three high-risk groups for whom no standard of care exists have been identified in the literature:

- Patients with *TP53* mutations ± del(17p), del(11q), or complex karyotypes (i.e., more than three cytogenetic aberrations)
- Patients with a short initial duration of remission (<24-36 months)
- Patients who have no response to therapy or progress on therapy

Del(17p), del(11q), and other genetic mutations have been identified as important independent predictors of disease progression and survival as shown in **Table 4** (Dohner, Stilgenbauer et al. 2000).

Table 4: Genetic mutations and disease outcomes in CLL

Abnormality	Percentage of patients	Median treatment-free interval (months)	Median OS (months)
Del(17)(p13.1)	7%	9	32
Del(11)(q22.3)	18%	13	79
Trisomy 12	16%	33	114
Del(13)(q14)	55%	49	133
None detected	18%	92	111

CLL = chronic lymphocytic leukemia
SOURCE: (Dohner, Stilgenbauer et al. 2000)

The survival rates among patients with CLL according to mutation status are shown in **Figure 2** in **Section 4.1.6.2**.

4.1.6 Prognostic factors

CLL is considered generally incurable, with limited treatment options. The prognosis of patients with CLL is dependent on a variety of factors that are patient-related (e.g., age, sex, comorbidities, performance status), disease-related (duration of remission, disease stage, cytogenetics, marrow failure, immunodeficiency, lymphomatous transformation, biomarkers), or treatment-related (type of treatment, response, toxicity, minimal residual disease status [MRD])-related factors (Oscier, Dearden et al. 2012). The 10-year relative survival rate is 65% for patients younger than 70 years of age and 55% for patients between 70 and 79 years of age (Brenner, Gondos et al. 2008). In comparison, the 5-year relative survival rate for patients with MCL, which has a worse prognosis than CLL, is 27% (Cancer Research UK 2013).

A comprehensive analysis of the German CLL Study Group database was undertaken to develop a prognostic index for patients with CLL that was then externally validated in an independent patient cohort of newly diagnosed patients with CLL using the Mayo Clinic CLL database (Pflug, Bahlo et al. 2013).

Table 5 shows the grading of adverse factors for patients with CLL using the German CLL Study Group prognostic index.

Table 5: Grading of adverse factors for patients with CLL

Variable	Definition	Grading*
Del(17p)	Present	6
B2M	>3.5 mg/L	2
	>1.7 mg/L to ≤3.5 mg/L	1
Thymidine kinase	<10.0U/L	2
Sex	Male	1
Del(11q)	Present	1
IGVH	Unmutated	1
Age	>60 years	1
ECOG	>0	1

* Higher grading scores indicate worse prognosis
B2M = β_2-microglobulin; CLL = chronic lymphocytic leukemia; ECOG = Eastern Cooperative Oncology Group; IGVH = immunoglobulin variable region heavy chain
SOURCE: (Pflug, Bahlo et al. 2013)

Table 6 shows the patient distribution in risk groups with 5-year OS rates. Higher index scores correspond to increased risk levels and decreased 5-year OS rates.

Table 6: Patient distribution in risk groups and 5-year OS rates

Index score	Risk group	Patients, % (TD/VD)	5-year OS, % (TD/VD)
0 to 2	Low risk	24.5/16.4	95.2/98.3
3 to 5	Intermediate risk	37.6/56.5	86.9/95.4
6 to 10	High risk	33.5/24.1	67.6/75.4
>10	Very high risk	4.3/3.0	18.7/18.7

CLL = chronic lymphocytic leukemia; TD/VD = training dataset/validation dataset; OS = overall survival
SOURCE: (Pflug, Bahlo et al. 2013)

The prognosis of patients with CLL is dependent on a variety of patient (age, gender, comorbidities, performance status), disease (disease stage, cytogenetics, marrow failure, immunodeficiency, lymphomatous transformation, biomarkers) and treatment (type of treatment, response, toxicity, minimal residual disease status MRD)-related factors (Oscier et al 2012); disease staging is the most important of the prognostic factors.

4.1.6.1 Disease staging

Staging systems are used to evaluate a patient at diagnosis based on the results of physical examination and blood tests in order to determine prognosis and decide on therapy (Eichhorst, Dreyling et al. 2011). There are two staging systems: the Binet system is most commonly used in Europe and the Rai system is used in the US (**Table 7**). Patients can be classified into low, intermediate, and high-risk groups on the basis of these staging systems.

Life expectancy depends on the stage at which the disease is diagnosed. Patients with Binet stage A disease generally survive for at least 10 years. For patients with Binet stage B, median survival is around 6.5 years (Yee and O'Brien 2006; Eichhorst, Dreyling et al. 2011). Binet stage C or Rai stage IV is the most advanced stage of CLL.

Table 7: Binet and Rai staging systems for CLL

Stage	Risk group	Features	Median survival
Binet stage			
A	Low	Hb ≥100 g/L or platelet count ≥100 x 10^9/L <3 lymphoid areas*	>10 years
B	Intermediate	Hb ≥100 g/L or platelet count ≥100 x 10^9/L ≥3 lymphoid areas	>8 years
C	High	Hb <100 g/L or platelet count <100 x 10^9/L	6.5 years
Rai stage			
0	Low	Lymphocytosis only	>10 years
I	Intermediate	Lymphocytosis and lymphadenopathy	>8 years
II		Lymphocytosis and hepatomegaly or splenomegaly with or without lymphadenopathy	
III/IV	High	Hb <110 g/L or platelet count <100 x 10^9/L	6.5 years

*The five lymphoid areas comprise unilateral or bilateral cervical, axillary, and inguinal lymphoid, hepatomegaly, and splenomegaly
CLL = chronic lymphocytic leukemia; Hb = hemoglobin
Adapted from: (Eichhorst, Dreyling et al. 2011; Oscier, Dearden et al. 2012)

4.1.6.2 Prognostic markers and cytogenetic abnormalities

Numerous factors have been identified and evaluated in patients with CLL, including serum markers (e.g., thymidine kinase, β_2-microglobulin), and genetic markers, such as IGVH mutational status and cytogenetic abnormalities detected by fluorescence in situ hybridization (FISH) (e.g., del(11q), del(17p), CD38 expression, and ZAP-70 expression) (Zelenetz, Gordon et al. 2014). Considerable variation exists in the use of prognostic markers and tests in the diagnostic workup for CLL, as shown in **Table 8** (NexGenvion Pharmaceuticals 2013), which shows the frequency of prognostic marker use in the US, EMEA, UK, France, Germany, Italy, and Spain. A breakdown of the use of prognostic markers by country or region is provided in **Appendix 7.3**.

Table 8: Use of prognostic markers and tests in the diagnostic workup for CLL

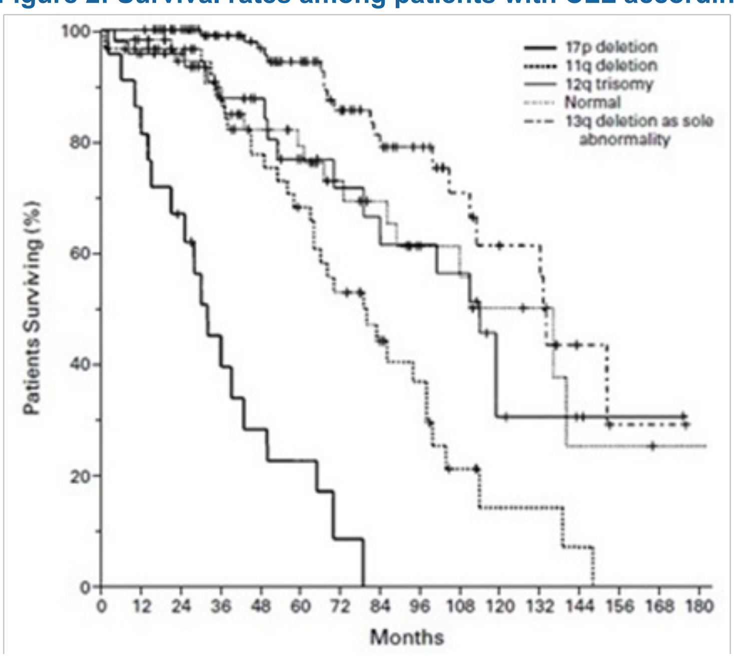
Marker	Frequency, %
B2M	46.8 to 94.1
CD38	50.0 to 78.4
FISH	58.0 to 84.3
IGVH	31.9 to 60.8
Lymphocyte doubling time	31.5 to 76.0
Thymidine kinase	2.1 to 47.1
ZAP70	21.3 to 64.0

B2M = β_2-microglobulin; CLL = chronic lymphocytic leukemia; FISH = fluorescence in situ hybridization; IGVH = immunoglobulin variable region heavy chain
SOURCE: (NexGenvion Pharmaceuticals 2013)

Treatment outcomes in CLL are strongly impacted by several molecular features. In particular, deletion of the short (p) arm of chromosome 17 (del(17p13.1)) results in the loss of *TP53* and has been identified as one of the poorest prognostic factors for CLL (Cuneo, Cavazzini et al. 2009; Zenz, Mertens et al. 2010), with a median life expectancy of less than 2 to 3 years from the time of initial diagnosis (Dohner, Stilgenbauer et al. 2000).

Relative to other chromosomal aberrations, del(17p13.1) is associated with shorter time to disease progression, shorter response duration, lack of response to therapy, and shorter overall survival (Dohner, Stilgenbauer et al. 2000; Hallek, Cheson et al. 2008; Zenz, Gribben et al. 2012). The survival rates among patients with CLL according to mutation status are shown in **Figure 2**.

Figure 2: Survival rates among patients with CLL according to mutation status



CLL = chronic lymphocytic leukemia
SOURCE: (Dohner, Stilgenbauer et al. 2000)
NOTE: Permission must be sought from the publisher before reproducing this figure for use with an external audience

Patients with CLL who relapse generally tend to accumulate genetic mutations, particularly del(17p) and del(11q). The more relapses patients incur, the more resistant they become to therapy (Maddocks and Lin 2009; Hillmen 2010). Del(17p) has a higher frequency in pretreated patients than in previously untreated patients, increasing during the course of the disease with approximately 30% to 50% of patients with relapsed or refractory disease having the deletion (Stilgenbauer and Zenz 2010).

- **Mutational status of IGVH, ZAP-70, CD38**

In CLL, approximately 50% of patients have a somatic hypermutation in their immunoglobulin heavy chain (IGVH) gene (Fais, Ghiotto et al. 1998). CLL with unmutated IGVH genes have an unfavorable prognosis: their long-term PFS and OS are worse compared with the outcomes for patients with the mutated IGVH (Rodriguez-Vicente, Diaz et al. 2013).

ZAP-70 is a protein tyrosine kinase that is involved in cell signaling following binding of an antigen by receptors on the lymphocyte. Normally, ZAP-70 is only expressed on T cells, but in CLL, it can be expressed in B cells; CD28 and ZAP-70 are functionally linked and are associated with poor outcomes (Deaglio, Vaisitti et al. 2007; Dreyling, Thieblemont et al. 2013). **Table 9** shows the impact of IGVH, ZAP70, and CD38 on CLL prognosis. Other mutations associated with short time to progression and survival are NOTCH1, SF3B1, BIRC3, and MYD88 (Wang, Lawrence et al. 2011; Foa, Del Giudice et al. 2013).

Table 9: Impact of IGVH, ZAP70, and CD38 on CLL prognosis

Marker	Frequency, %	Time to treatment (months)	OS (months)
IGVH			
Mutated	47	110	300
Unmutated	53	42	115
ZAP70			
Negative	54	110	NR
Positive	46	35	NR
CD38			
Negative	67	94	193
Positive	33	40	109

CLL = chronic lymphocytic leukemia; OS = overall survival
SOURCE: (Rassenti, Huynh et al. 2004; Rassenti, Jain et al. 2008)

- **Other prognostic factors**

Raised ß$_2$-microglobulin (B2M) was also correlated with reduced PFS and OS in a clinical trial of alkylating agent and purine analogue treatment (Oscier, Wade et al. 2010).

4.2 Burden of disease in CLL

As described in **Section 4.1.3**, the clinical sequelae of CLL can have substantial negative impacts on patients' quality of life (QoL) as a result of disease-related symptoms such as fatigue, recurrent infections, and anemia, treatment-related AEs, and the psychological, socioeconomic and functional effects of living with the disease. The emotional well-being of patients with CLL has been shown to be significantly lower than that of the general population, as well as patients with other types of cancer (P=0.001) (Shanafelt, Bowen et al. 2007).

CLL is a life-threatening disease due to development of cytopenias and impaired production of normal immunoglobulin. Patients with CLL have a high risk of morbidity and mortality from disease-related infections, which are commonly due to bacteria and influenced by the degree of hypogammaglobulinemia. The frequency of serious infections requiring hospitalization for intravenous antibiotics in patients with fludarabine-refractory CLL/SLL is extremely high (Perkins, Flynn et al. 2002). In more advanced stages of disease, neutropenia due to bone marrow infiltration and/or cytotoxic therapy may also contribute to the increased risk of infection-related morbidity and mortality. Defects in cell-mediated immunity also appear to be a predisposing factor to infections in patients treated with newer purine analogs (Wadhwa and Morrison 2006).

4.2.1 Impact on quality of life

Historically, there has been a limited focus on the QoL impact of this disease on patients (Stephens, Gramegna et al. 2005) (see **Appendix 7.1**). However, long-term data regarding the impact of treatment on QoL are important in order to understand the holistic impacts of treatments on patients (Zent 2012), especially considering the chronic nature of CLL. As such, addressing and improving patients' QoL over the long term is a major goal in the initial phase of treatment (Zent 2012).

While there is a paucity of data on the QoL impact of relapsed or refractory CLL, currently available patient-reported data suggest that CLL has the greatest impact on emotional domains of patient QoL. Shanafelt and colleagues evaluated fatigue and QoL in CLL, surveying a total of 1,482 previously treated patients, untreated patients, and patients currently on active treatment. The physical, social, functional well-being, and overall QoL scores in the Shanafelt study were better than published population norms, but the mean emotional well-being score in all included patients with CLL was clinically meaningfully lower than that of the general population and of patients with other types of cancers (Brucker, Yost et al. 2005; Shanafelt, Bowen et al. 2007). In the same study, patients with CLL reported statistically significantly higher rates of fatigue than the general population; patient-reported fatigue increased with disease stage (Shanafelt, Bowen et al. 2007).

Recent progress has been made in understanding the effect of treatment on the QoL of patients with CLL. In a recently published study of 777 previously untreated patients randomized to receive first-line treatment with either chlorambucil, fludarabine monotherapy, or FC, Else and colleagues utilized the EORTC-QLQC30 to assess the impact of CLL on patients over a 5-year period spanning 1999 and 2004 (Else, Cocks et al. 2012). Results of the study, which included 136 participating centers (mostly in the UK), indicated that patients greater than 70 years of age reported statistically significantly worse scores in physical functioning compared with younger patients (Else, Cocks et al. 2012). Older patients also reported significantly lower scores in the role functioning domain (mean 63 points *vs* 73 points; 95% CI, 69 to 77; *P*=0.006) and in the global HRQoL score (mean 54 points *vs* 67 points; 95% CI, 48 to 59; *P*=0.00001) (Else, Cocks et al. 2012). Although the authors do not explore the factors involved in the worse scores reported in older patients, comorbidities are more common in this population, with around 50% having at least one major accompanying disease (Klepin, Rizzieri et al. 2013), which may have contributed to the lower functioning and QoL scores.

Other factors that impacted patient-reported QoL included disease response to treatment and experience of side effects. Patients who were non-responders and patients whose disease had progressed reported worse QoL than responders and those patients whose disease remained in the first remission (Else, Cocks et al. 2012). Else and colleagues found that fatigue, leading to impaired role functioning, may be a key factor influencing QoL in CLL. Their findings were supported by the strong correlation throughout the study period between patients' overall (global) assessment of their HRQoL and both role functioning and fatigue. The experience of nausea, vomiting, sleep disturbance, constipation, and diarrhea, which showed the least association with the overall patient-reported global QoL score, also contribute to the negative impacts of the disease on patient QoL (Else, Cocks et al. 2012).

4.2.2 Economic burden of disease

Although CLL is the most common form of adult leukemia, literature on its economic burden is limited, and there are no published studies to date which examine the economic burden of relapsed or refractory disease (see **Appendix 7.1**). The most recent systematic literature review of the economic burden in CLL included 13 studies from 1990 to 2002 (Stephens 2005). The majority of studies were cost-identification or cost-comparison studies, and five were cost-effectiveness studies of medical interventions to treat patients with CLL. The results of the studies are not included in this GVD due to the age of the studies (1990 to 2002), which limits the value of this literature review and warrants an assessment of more recent literature.

In a study of the disease burden of CLL in the European Union using GLOBOCAN 2002 data, Watson and colleagues highlighted the scant epidemiological and cost of illness data available for this disease. From their evaluation of epidemiological data gathered from global and EU cancer databases, the authors concluded that the number of individuals burdened by the disease within the European Union is considerable and thousands of patients require treatment and physician care, which has cost implications for member states (Watson, Wyld et al. 2008).

A retrospective database analysis conducted by Blankart and colleagues of 7.6 million individuals enrolled in a large German sickness fund offers more recent insight into the total costs associated with CLL treatment (Blankart, Koch et al. 2013). As shown by the study, CLL imposes a high economic burden on sickness funds and society. In order to determine the cost attributed to CLL, the authors used a case-control design with a randomly selected control group of 150 individuals matched by age and sex with the 4198 enrolled patients with CLL. The authors created a control group consisting of 4198 × 150 observations and used the age- and sex-specific controls multiple times if one age and sex group contained more than one prevalent subject. The costs attributable to CLL were calculated by comparing the differences between the two groups. Healthcare utilization was significantly different between the patients with CLL and the control cohort (**Table 10**). Patients with CLL experienced significantly more sick leave days (P<0.001) and physician contacts (P<0.001), and required more prescription drugs (P<0.001) than the control cohort.

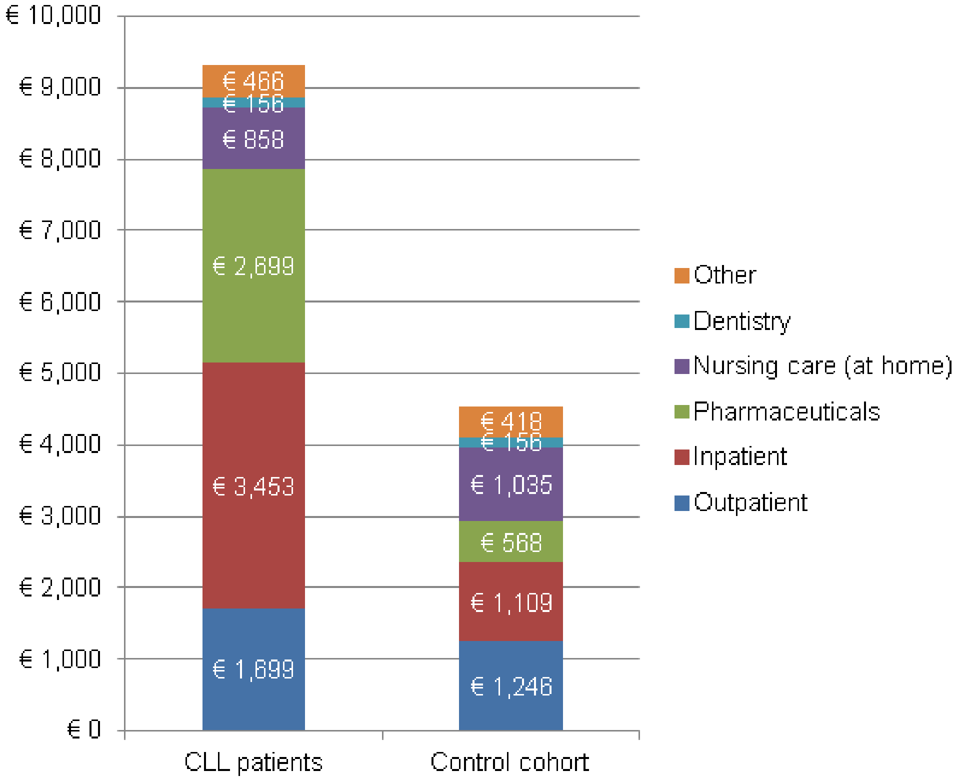
Table 10: Healthcare utilization in patients with CLL and matched controls

Parameter	Patients with CLL Mean (SD)	Controls Mean (SD)	*P*
Sick leave days (paid and unpaid)	7.6 (35.5)	3.1 (19.6)	<0.001
Number of physician contacts	14.7 (7.5)	9.4 (6.6)	<0.001
Number of inpatient stays	0.95 (2.10)	0.36 (0.96)	0.007
Average length (if hospitalized), days	9.4	8.8	—
Number of prescriptions	18.2 (19.3)	12.8 (14.4)	<0.001

CLL = chronic lymphocytic leukemia; SD = standard deviation
SOURCE: (Blankart, Koch et al. 2013)

The study evaluated healthcare utilization in patients with CLL regardless of the line of treatment and did not account for differences between patients in the first-line or relapsed or refractory settings. As shown in **Figure 3**, the total average direct medical costs for a patient with CLL in Germany were € 9,331 per year compared with € 4,533 for the control cohort (P<0.001) (Blankart, Koch et al. 2013). The majority of costs for patients with CLL were due to the cost of inpatient care (€ 3,453) and drug costs (€ 2,699). The majority of drug costs were related to the cost of chemotherapy agents (€ 835), cytostatic drugs (€ 504), and anti-infective drugs for systemic use (€ 517). When accounting for the prevalence of CLL, the authors estimated the overall economic burden of CLL to be € 201.1 million per year (CI € 187.6 million to € 223.6 million) for the German sickness fund.

Figure 3: Average direct medical costs per person per year in Germany



CLL = chronic lymphocytic leukemia
SOURCE: (Blankart, Koch et al. 2013)

4.3 Current management of relapsed or refractory CLL

4.3.1 Overview of CLL treatment and goals of therapy

Patients with asymptomatic, early-stage disease should be monitored without undergoing therapy unless there is evidence of disease progression; a number of studies have indicated that the use of alkylating agents in patients with early-stage disease does not prolong survival (Hallek, Cheson et al. 2008). According to the International Workshop on Chronic Lymphocytic Leukemia (IWCLL) recommendations, patients should have clearly documented active disease before initiating therapy (**Table 11**).

CLL principally remains incurable with chemoimmunotherapy, with the natural history of the disease one of repeated relapse. Some patients follow an aggressive course from the outset, which is associated with poor treatment outcomes (see **Section 4.1.5** and **Section 4.1.6**). Throughout treatment, the goal of therapy is to relieve symptoms, which can lead to an improvement in QoL. The goals of therapy in CLL may also depend on individual patient characteristics. For physically fit patients, the goal may be increased overall survival, but for patients with reduced physical fitness, the time to progression or HRQoL may be more suitable goals (Hallek, Cheson et al. 2008).

Table 11: IWCLL treatment recommendations in CLL

IWCLL	Recommendation
Primary treatment decisions	While patients at intermediate (stages I and II) and high risk (stages III and IV) usually benefit from initiation of treatment, some of these patients can be monitored without therapy until they have evidence of progressive or symptomatic disease. Active disease should be clearly documented for protocol therapy. At least 1 of the following criteria should be met: 1. Evidence of progressive marrow failure as manifested by the development of, or worsening of, anemia and/or thrombocytopenia 2. Massive (i.e., ≥6 cm below the left costal margin) or progressive or symptomatic splenomegaly 3. Massive nodes (i.e., ≥10 cm in longest diameter) or progressive or symptomatic lymphadenopathy 4. Progressive lymphocytosis with an increase of more than 50% over a 2-month period or LDT of <6 months 5. Autoimmune anemia and/or thrombocytopenia that is poorly responsive to corticosteroids or other standard therapy 6. Constitutional symptoms, defined as any 1 or more of the following disease-related symptoms or signs: a. Unintentional weight loss of ≥10% within the previous 6 months b. Significant fatigue (i.e., ECOG PS 2 or worse; inability to work or perform usual activities) c. Fevers higher than 100.5° F or 38.0° C for ≥2 weeks without other evidence of infection; or Night sweats for more than 1 month without evidence of infection
Second-line treatment decisions	Second-line treatment decisions should follow the same indication as those used for initiation of first-line treatment. Patients who have resistant disease, a short TTP after the first treatment, and/or leukemia cells with del 17p often do not respond to standard chemotherapy and have a relatively short survival. Therefore, such patients should be offered investigative clinical protocols, including allogeneic HSCT.

ECOG PS = Eastern Cooperative Oncology Group performance status; HSCT = hematopoietic stem cell transplant; IWCLL = International Workshop on Chronic Lymphocytic Leukemia; LDT = lymphocyte doubling time; TTP = time to progression
SOURCE: (Hallek, Cheson et al. 2008)

Standard therapies in CLL include chemoimmunotherapy (e.g., FCR, BR) in first-line treatment. Choice of treatment at relapse depends on the quality and duration of response to previous therapy, age, performance status, presence of cytogenetic abnormalities such as the development of del(17p), and treatment intent (i.e., aggressive or palliative) (Gribben and O'Brien 2011). Options at first relapse include repeating front-line therapy or switching to an alternative regimen. In those patients relapsing after initial chemoimmunotherapy, patients are commonly re-treated with a second chemoimmunotherapy regimen if their initial duration of response was more than 24 to 36 months (Eichhorst, Dreyling et al. 2011). At second or later relapses in which there is often a shorter duration of response, a palliative approach may be considered or new therapeutic strategies employed (Eichhorst, Dreyling et al. 2011).

Patients who have resistant disease, a short duration of response after the first treatment, and/or presence of del(17p) often do not respond to standard chemotherapy or chemoimmunotherapy and have a relatively short survival. It is recommended by the IWCLL treatment guidelines that such patients should be offered investigative clinical protocols, including allo-SCT (Hallek, Cheson et al. 2008). The optimal role of stem cell transplantation in CLL is still to be defined (Gribben and O'Brien 2011). Many patients are ineligible because of age or comorbidities; the European Society for Blood and Marrow Transplantation (EBMT) recommends allo-SCT in eligible CLL patients with p53 abnormalities (e.g., del(17p)) and in younger patients who fail to respond, or relapse within 24 months of front-line chemoimmunotherapy (Ghielmini, Vitolo et al. 2013).

CALLOUT: The CLL8 trial

CLL8 was a landmark phase 3 trial of fludarabine/rituximab-based therapy in CLL conducted by the German CLL Study Group (G-CLLSG) (Schnaiter and Stilgenbauer 2011; Jain and O'Brien 2013).

A total of 817 patients were randomized between FCR (*n*=408) and FC (*n*=409) (Jain and O'Brien 2013). FCR was shown to be superior to FR across all outcomes including ORR (90% *vs* 80%), CR rate (44% *vs* 22%), PFS (median 51.8 months *vs* 32.8 months), and 3-year OS (87% *vs* 83%). At 3 years, 65% of patients in the FCR arm were free of progression compared with 45% in the FC arm (45%). MRD negativity was also significantly more common in the FCR arm (48%) *vs* FC (27%).The toxicity associated with FCR included grade 3/4 neutropenia in 34% of cases.

Impact of (del17p)

In CLL8, patients with del(17p) had inferior outcomes with both fludarabine-based regimens as compared with other cytogenetic risk categories (Jain and O'Brien 2013). The 3-year PFS for patients with del(17p) was 0% on FR and 18% on FCR (Schnaiter and Stilgenbauer 2011). The 3-year OS rates were also significantly reduced in del(17p) patients treated with FCR (38%) and FC (37%) (Schnaiter and Stilgenbauer 2011). These results highlight the lack of efficacy of standard front-line fludarabine-based therapy in CLL patients with the del(17p) mutation – or those who acquire it during the evolution of their disease.

CLL8: Results for Patients with del(17p)				
Regimen	**ORR**	**Median PFS**	**3-yr PFS**	**3-yr OS**
FCR	68%	11.3 months	18%	38%
FC	34%	6.5 months	0%	37%

CLL = chronic lymphocytic leukemia; FC = fludarabine, cyclophosphamide; FCR = fludarabine, cyclophosphamide, rituximab; ORR = overall response rate; OS = overall survival; PFS = progression-free survival;

4.3.2 Treatment pathway

Treatment decisions for newly diagnosed patients can be guided by a combination of factors, including the patient's staging, fitness, and mutation status as shown by the proposed treatment algorithm in **Table 12**. (Staging and mutation status are discussed in **Section 4.1.6**.) "Go go" is used to describe patients who are fit for treatment, "no go" for patients who are unfit, and "slow go" for patients who fall between these two extremes (Eichhorst, Goede et al. 2009; Leblond 2012).

Allo-SCT, which is the only potentially curative therapy for CLL, is limited to patients with an appropriate matched donor who are fit for aggressive therapy. Allo-SCT is less commonly used in clinical practice unless other therapeutic options have been exhausted. The percentage of patients with CLL who undergo allo-SCT worldwide is 11% (Gratwohl, Baldomero et al. 2013). Older registry analyses have reported non-relapse mortality rates of up to 44% for allo-SCT in CLL. Chronic graft versus host disease (cGVHD) is a common complication of allo-SCT, occurring in approximately 40% to 70% of cases (Socie, Salooja et al. 2003). Infection is the primary cause of non-relapse mortality related to cGVHD. Reduced-intensity conditioning regimens for allo-SCT have decreased the non-relapse mortality rates to between 15% and 25% (Dreger 2009). Depending on the immunomodulatory strategy employed, extensive cGVHD develops in up to 60% of patients at risk after reduced intensity conditioning (RIC) allo-SCT for CLL (Dreger 2009).

A study by Fraser and colleagues (2006) documented a significant reduction in self-reported long-term health status in patients who received allografts for various hematological malignancies, and who had active cGVHD. Of patients with a history of active cGVHD, 81.3% had an adverse outcome in at least one domain (general health, mental health, functional impairment, activity limitation, pain, anxiety), compared with 40.2% of patients with resolved cGVHD, and 42.1% of subjects with no history of cGVHD (Fraser, Bhatia et al. 2006).

Table 12: Proposed algorithm for first-line treatment of CLL

Patients	Fitness	Treatment
"Go go"	• Medically fit • No or mild comorbidities • Normal life expectancy	Intensive chemotherapy, preferably in a clinical trial
"Slow go"	• Medically less fit • Multiple or severe comorbidities • Unknown life expectancy	Clinical trial for patients with comorbidities, or therapy adapted to patient's comorbidity burden and individual risk level
"No go"	• Medically frail • Fatal comorbidity • Reduced life expectancy	No chemotherapy

CLL = chronic lymphocytic leukemia
SOURCE: (Eichhorst, Goede et al. 2009)

4.3.2.1 Treatment guidelines for CLL

Evidence-based guidelines for first-line therapy and relapsed or refractory CLL are provided by ESMO, NCCN, and Società Italiana di Ematologia/Società Italiana di Ematologia Sperimentale/Gruppo Italiano Trapianto di Midollo Osseo (SIE/SIES/GITMO), along with other local country guidelines (Eichhorst, Dreyling et al. 2011; Kater, Wittebol et al. 2011; DGHO 2012; Mauro, Bandini et al. 2012; Oscier, Dearden et al. 2012; Aurrant, Callet-Bauchu et al. 2013; Zelenetz, Gordon et al. 2014). A comparison of guidelines for relapsed or refractory CLL is shown in **Appendix 7.4**.

4.3.2.2 Treatment selection in relapsed or refractory CLL

The choice of therapy in patients with relapsed or refractory disease depends on a variety of factors including (Veliz and Pinilla-Ibarz 2012):

- patient age
- performance status
- level of fitness (e.g., biological fitness, bone marrow capacity, and comorbidities)
- cytogenetic abnormalities (see **Section 4.1.6.2** and **Appendix 7.3**)
- DOR to initial therapy
- type of prior therapy
- disease-related manifestations.

An additional factor in treatment selection is the consideration of treatment-related complications. The most common cause of morbidity and mortality in patients with CLL is infection, primarily due to hypogammaglobulinemia (Dearden 2008; Desai and Pinilla-Ibarz 2012). Treatment with immunosuppressive therapies, such as purine analogues and monoclonal antibodies, increases the risk of infection over time in patients with CLL, especially in heavily pretreated patients with active disease (Dearden 2008).

CLL is also frequently associated with autoimmune phenomena, the most common of which is autoimmune hemolytic anemia (AHA); immune thrombocytopenia purpura and pure red blood cell aplasia are observed less frequently (Dearden 2008; Hodgson, Ferrer et al. 2011). AHA has also been shown to occur more often in patients receiving treatment with chlorambucil or fludarabine monotherapy than with the combination of fludarabine and cyclophosphamide (Dearden, Wade et al. 2008).

ESMO and NCCN guidelines for supportive care in CLL include recommendations for the management of recurrent sino-pulmonary infections, anti-infective prophylaxis, autoimmune cytopenia, vaccination, blood product support, tumor lysis syndrome (TLS), tumor flare reactions, and thromboprophylaxis (Eichhorst, Dreyling et al. 2011; Zelenetz, Gordon et al. 2014).

Numerous drug-related toxicities can result from CLL therapies (O'Brien 2014). Treatment with rituximab has resulted in symptomatic hypogammaglobulinemia that has necessitated intravenous immunoglobulin (IVIG) administration. Other drug toxicities associated with rituximab include increased incidence of grade 3 and 4 infections, neutropenia, hepatitis B reactivation, squamous cell skin carcinoma, and progressive multifocal leukoencephalopathy (Casulo, Maragulia et al. 2013). Ofatumumab and alemtuzumab can cause serious AEs, such as infusion-related reactions, HBV reactivation, and progressive multifocal leukoencephalopathy (Wierda, Padmanabhan et al. 2011; Skoetz, Bauer et al. 2012). AEs associated with lenalidomide include tumor flare reactions, tumor lysis syndrome (TLS), and myelosuppression (Chanan-Khan, Miller et al. 2006; Moutouh-de Parseval, Weiss et al. 2007).

The long-term development of second primary malignancies attributed to chemotherapy can be another cause of CLL morbidity and mortality (Ricci, Tedeschi et al. 2011; Zhou, Tang et al. 2012). Additional AEs associated with other agents evaluated in the treatment of relapsed or refractory CLL can be found in **Table 13** of **Section 4.4**.

4.4 Overview of unmet need

CLL is generally incurable (a small proportion of patients who are fit for aggressive therapy may be cured by allo-SCT), and treatment will alleviate signs and symptoms but relapse is expected and inevitable (Kojima, Duvvuri et al. 2012). Additional lines of therapy often provide shorter PFS in patients with relapsed CLL compared with first-line treatment.

4.4.1 Unmet need in high-risk CLL groups

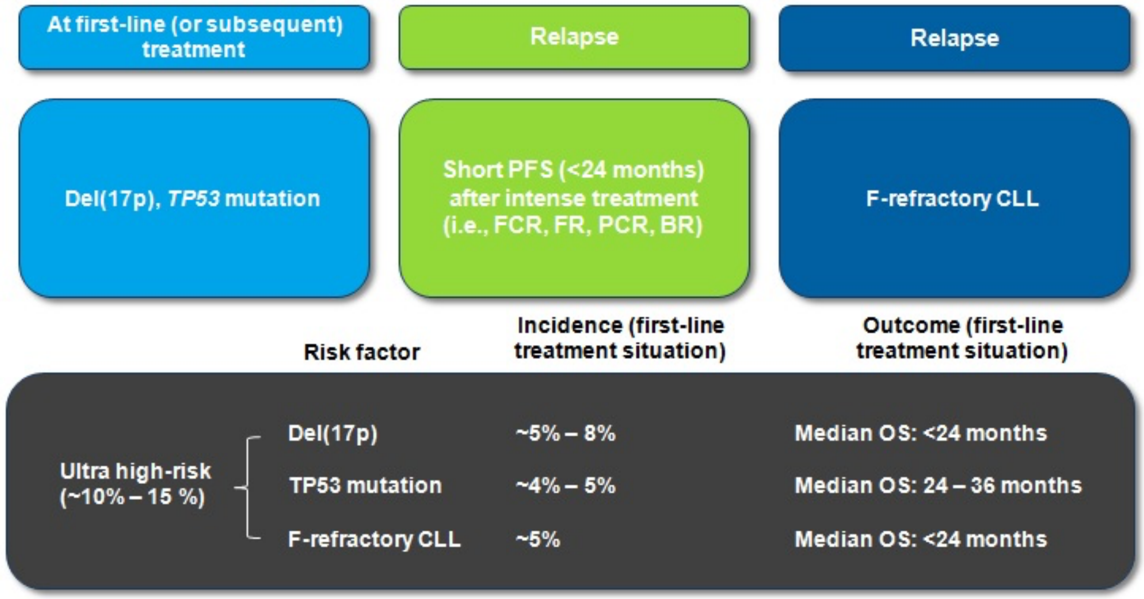
Current treatment options for patients with relapsed or refractory CLL are poor, particularly for the following high-risk groups for whom no standard of care exists:

- Patients with *TP53* mutations ± del(17p), del(11q), or complex karyotypes (i.e., more than three cytogenetic aberrations)
- Patients with a short initial duration of remission (<24-36 months)
- Patients who have no response to therapy or progress on therapy

A clear need exists for new agents to provide valuable treatment alternatives with reduced risk of treatment-related AEs for patients in high-risk CLL groups and for elderly patients with comorbidities who are unable to tolerate chemoimmunotherapy.

Stilgenbauer and colleagues identified patients with del(17p), the *TP53* mutation, and fludarabine-refractory CLL, as well as patients with suboptimal response to intense treatment (e.g., FCR), as an ultra high-risk group; the median life expectancy of these patients is below 2 to 3 years with standard regimens (**Figure 4**) (Stilgenbauer and Zenz 2010).

Figure 4: Definition of ultra high-risk CLL

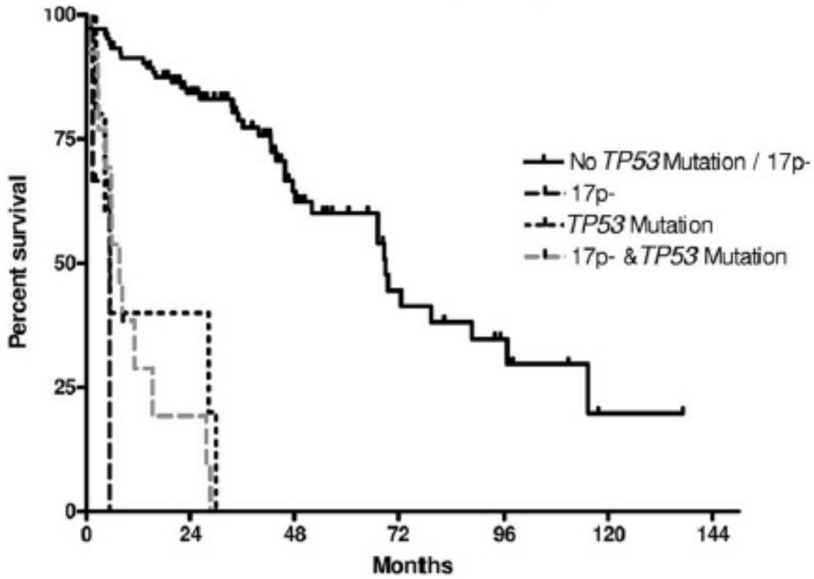


BR = bendamustine plus rituximab; CLL = chronic lymphocytic leukemia; FC = fludarabine, cyclophosphamide, rituximab; FR = fludarabine, rituximab; F-refractory = fludarabine-refractory; OS = overall survival; PCR = pentostatin, cyclophosphamide, rituximab
SOURCE: (Stilgenbauer and Zenz 2010)
NOTE: Permission must be sought from the publisher before reproducing this figure for use with an external audience

In the first-line setting, ultra high-risk patients with *TP53* mutations, del(17p), and/or fludarabine-refractory CLL represent 10% to 15% of all patients with CLL and have an estimated OS of less than 3 years, significantly shorter than the almost 7-year survival seen in those without these high-risk clinical features. Importantly, chemoimmunotherapy appears to select for those poor-risk CLL clones, with the incidence of *TP53* mutations and del(17p) increasing to approximately 20% at first relapses, and as much as 40% to 50% in fludarabine-refractory CLL. Another high-risk group of patients are those with a short initial duration of remission (<24-36 months), with OS of less than 2 years following initial treatment failure. Both of these high-risk patient populations (*TP53*/del(17p)) and short first remission of <24-36 months) have a high unmet medical need, and new treatment approaches are critically needed (Stilgenbauer and Zenz 2010).

The presence of any *TP53*/del(17p) abnormality (i.e., deletion with mutation, deletion without mutation, mutation without deletion) is associated with inferior survival as shown in **Figure 5**. The median OS (from the time of study) was shown to be inferior in groups with del(17p) and *TP53* mutation (7.6 months), sole *TP53* mutation (5.5 months), and del(17p) without *TP53* mutation (5.4 months), compared with patients without *TP53* mutation or del(17p) (69 months; *P*<0.001) (Zenz, Kröber et al. 2008).

Figure 5: Overall survival of patients with CLL according to absence or presence of *TP53* mutation/17p deletion (*n*=125; *P*<0.001)



CLL = chronic lymphocytic leukemia
SOURCE: (Zenz, Kröber et al. 2008)
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4.4.2 Treatment challenges

Treatment of patients with relapsed or refractory CLL is challenging due to the low rates of response associated with many existing therapies; few patients, especially those in high-risk groups, are able to achieve long durable benefit with PFS greater than 2 years from available single-agent or combination therapies. In addition to having substantial toxicity, current regimens for patients with CLL generally become less effective with recurrent treatment. Palliative chemotherapy further depletes the bone marrow reserve and the ability to reconstitute normal immunity, while continuing to precipitate states of thrombocytopenia, neutropenia, and anemia, which require additional supportive measures. The long-term development of second primary malignancies has also been attributed to chemotherapy and is an important consideration in treatment selection (Ricci, Tedeschi et al. 2011; Zhou, Tang et al. 2012). New therapeutic agents are still urgently needed for patients with relapsed or refractory CLL who do not respond to currently available treatments or cannot achieve a sustained response (Barrientos and Rai 2013).

Studies of treatment with single-agent therapies in patients with relapsed or refractory CLL reported the following outcomes (Sandoz [US PI] 2010; Wierda, Kipps et al. 2010; Genzyme [SmPC] 2012; Niederle, Megdenberg et al. 2013) (see **Table 13**):

- ORRs of 21% to 33% with alemtuzumab, 48% with fludarabine, 51% with ofatumumab, and 76% with bendamustine*
- Median PFS of 4 to 7 months with alemtuzumab, 5.5 months with ofatumumab, and 20.1 months with bendamustine*
- Median OS of 14.2 months with ofatumumab, 16 to 28 months with alemtuzumab, and 43.8 months with bendamustine*

Studies of combination regimens in patients with relapsed or refractory CLL reported outcomes as follows (Robak, Dmoszynska et al. 2010; Badoux, Keating et al. 2011; Fischer, Cramer et al. 2011; Badoux, Keating et al. 2013) (see **Table 13**):

- ORRs of 58% to 74% with FC, 59% with BR, 66% with lenalidomide plus rituximab, and 70% with FCR
- Median PFS of 15.2 months with BR, 21 months with FC, and 30.6 months with FCR
- Median OS of 33.9 months with BR, 46.5 months with FC, and 52 months with FCR

CLL also has highly variable disease presentations and cytogenetics, with life expectancies ranging from months to years (Schultz, Delioukina et al. 2011). Most patients with high-risk disease do not achieve sustained remission with existing therapies. Patients with *TP53* loss and/or mutation have a significantly lower response rate and short PFS and OS when treated with an alkylating agent, purine analog (fludarabine-based regimens), bendamustine, and mitoxantrone and rituximab, alone or in combination. In contrast, *TP53* status has a much lower effect on the response of patients treated with agents such as alemtuzumab. However, alemtuzumab was shown to have a short time to progression (TTP); 9.5 months in a single-arm trial (Keating, Flinn et al. 2002), and is associated with immune toxicity (Demko, Summers et al. 2008).

Treatment options are particularly scarce in patients with relapsed or refractory CLL with del(17p), as resistance to conventional chemotherapies is common in this population (Veliz and Pinilla-Ibarz 2012). In patients with CLL with del(17p), treatment with ofatumumab (*n*=27) achieved an ORR of 37% and a median PFS of 3.3 months (Wierda, Kipps et al. 2010). Combination regimens achieved the following efficacy in patients with CLL with del(17p) (Badoux, Keating et al. 2011; Fischer, Cramer et al. 2011; Badoux, Keating et al. 2013):

- ORRs of 7% with BR, 35% with FC, and 53% with lenalidomide plus rituximab
- Median PFS of 5 months with FC and 6.8 months with BR
- Median OS of 10.5 months with FC and 16.3 months with BR

There is a particular need for a licensed, efficacious and well tolerated treatment for patients with del(17p) given that as of August 2012, Genzyme Europe B.V. has withdrawn the license for alemtuzumab in patients with CLL for whom fludarabine combination therapy is not appropriate (EMA 2013). Ofatumumab

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* The efficacy of bendamustine monotherapy was evaluated in a study of 49 patients with CLL requiring treatment after one previous systemic regimen. None of the patients in the study were reported to have del(17p). In previously untreated CLL, bendamustine demonstrated an ORR of 59% and a median PFS of 18 months. OS was not reported. Teva Pharmaceuticals [US PI] (2013). Treanda [package insert]. North Wales, PA.

is the treatment of choice for patients who do not respond to alemtuzumab; however, it is not recommended by NICE or the SMC due to the uncertain efficacy and cost-effectiveness versus best supportive care. In addition, it is associated with short median PFS of 5.5 months for patients refractory to fludarabine and alemtuzumab, and a PFS of 3.3 months for those with del(17p) (Wierda, Kipps et al. 2010).

Furthermore, CLL is considered primarily a disease of the elderly, thus many patients are unable to tolerate aggressive regimens (Barrientos and Rai 2013) (see **Section 4.3.2.2**). In elderly patients with CLL, aggressive chemotherapy regimens often cannot be considered due to impaired performance status. More effective treatments with fewer toxicities are required for elderly populations who would otherwise be unable to tolerate standard regimens (**Table 13**) (Barrientos and Rai 2013; O'Brien, Furman et al. 2014).

Table 13 outlines the current treatments considered for relapsed or refractory CLL, as well as their characteristics, efficacy, safety, guideline recommendation, and registration status. Agents in this table were chosen based on ESMO, NCCN, and SIE/SIES/GITMO guidelines. With the exception of lenalidomide and chlorambucil, there is a lack of orally administered agents for this indication.

Table 13: Regimens used in relapsed or refractory CLL

Study design and population	ORR (%)	CR (%)	Median PFS (mo)	Median OS (mo)	Adverse events	Dosage form MOA	Registration status	Guideline recommendation†
Approved single-agent therapies and their use in combination regimens								
Fludarabine								
Fludarabine in CLL refractory to ≥1 prior alkylating-agent containing regimen (*N*=48) • Median age = NA (Sandoz [US PI] 2010)	48	13	NR	NR	Serious AEs >10%: Not reported	Injection Purine analog	FDA-approved as monotherapy in patients with CLL who have not responded or progressed following ≥1 alkylating agent regimen; and as FCR for previously untreated and previously treated CD20+ CLL EMA-approved as first-line treatment in CLL	ESMO III-B; NCCN 2A; SIE/SIES/GITMO
FC ± R in previously treated CD20+ CLL (*N*=552) • Median age = 62.0 years (FC arm); 63.0 years (FCR arm) • Unmutated IGVH: 65% (FC arm); 61% (FCR arm) • Alkylator refractory: 26% (FC arm); 27% (FCR arm) (Robak, Dmoszynska et al. 2010)	FC arm (*n*=276) 58.0 FCR arm (*n*=276) 69.9	 13.0 24.3	 20.6 30.6	 52 NR	Serious AEs >10%: grade 3 or 4 AEs: FC arm: neutropenia (40%) febrile neutropenia (12%) anemia (13%) FCR arm: neutropenia (42%) febrile neutropenia (12%) anemia (12%) thrombocytopenia (11%)			
FC in R/R CLL (*N*=284) including del(17p) (*n*=20) • Median age = 60 years • Median of 2 prior treatments (Badoux, Keating et al. 2011)	Total population 74 Del(17p) 35	 30 0	 21 5	 46.5 10.5	Serious AEs >10%: Not reported			
Bendamustine								
Bendamustine in relapsed CLL (*N*=49) • 1 prior systemic regimen • Median age = 68 years • Binet stage A or B: 45% • Del(17p) excluded (Niederle, Megdenberg et al. 2013)	76	27	20.1	43.8	Serious AEs >10%: neutropenia (20%) leukopenia (18%) infection (13%)	Injection Purine antagonist	Bendamustine: FDA-approved as monotherapy for CLL Rituximab: FDA-approved as part of FCR therapy for previously untreated and previously treated CD20+ CLL	ESMO III-B; NCCN 2A

Study design and population	ORR (%)	CR (%)	Median PFS (mo)	Median OS (mo)	Adverse events	Dosage form MOA	Registration status	Guideline recommendation[†]
BR in previously treated R/R CLL (*N*=78), including patients with del(17p) (*n*=14) • Median age = 66.5 years • Median of 2 prior treatments (Fischer, Cramer et al. 2011)	Total population				Serious AEs >10%: grade 3 or 4 AEs: neutropenia (23.1%) leukopenia (18%) thrombo-cytopenia (28.2%) anemia (16.6%) infections (12.8%)			
	59	9	15.2	33.9				
	Del(17p)							
	7	7	6.8	16.3				
Ofatumumab								
FA-ref CLL (*N*=95), including patients with del(17p) (*n*=27) • Median age = 64 years • Binet stage A or B: 49% (Wierda, Kipps et al. 2010)	FA-ref				Grade ≥3 AEs in ≥5% of all patients: infections (24%) neutropenia (12%) anemia (5%)	Injection Anti-CD20 monoclonal antibody	FDA- and EMA-approved as monotherapy for FA-ref CLL	NCCN 2A
	51	0	5.5	14.2				
	Del(17p)							
	37	0	3.3	NA				
Chlorambucil								
	NA	NA	NA	NA	NA	Oral Alkylating agent	FDA-approved for CLL treatment	
Other available therapies								
Lenalidomide								
LR in R/R CLL (Total population: *N*=59 and del(17p): *n*=15) • Median age = 62 years • Median of 2 prior treatments (Badoux, Keating et al. 2013)	Total population: 66; del(17p): 53	12	NA	NR, estimated survival at 36 mo: 71%	Serious AEs >10%: most common grade 3 or 4 AEs: neutropenia (73%) thrombocytopenia (34%) anemia (15%) pneumonia/ bronchitis (10%) neutropenic fever (10%) fatigue (14%)	Oral Immuno-modulating agent	Investigational	NCCN 2A

Study design and population	ORR (%)	CR (%)	Median PFS (mo)	Median OS (mo)	Adverse events	Dosage form MOA	Registration status	Guideline recommendation†
Alemtuzumab								
Alemtuzumab in patients with B-cell CLL who had received an alkylating agent and had failed fludarabine (*N*=93) • Median age = 66 years • Median of 3 prior treatments (Genzyme [SmPC] 2012)	33	2	4	16	NA	Injection Anti-CD52 monoclonal antibody	CLL license withdrawn[a]	ESMO III-B; NCCN 2A; SIE/SIES/GITMO
Alemtuzumab in patients with B-cell CLL pts who had failed to respond or relapsed following treatment with conventional chemotherapy (*N*=32) • Median age = 57 years • Median of 3 prior treatments (Genzyme [SmPC] 2012)	21	0	5	26				
Alemtuzumab in patients with B-cell CLL who had failed to respond or relapsed following treatment with fludarabine (*N*=24) • Median age = 62 years • Median of 3 prior treatments (Genzyme [SmPC] 2012)	29	0	7	28				
Idelalisib plus rituximab								
Idelalisib plus rituximab in patients with previously treated CLL (*N*=110) • Median age = 71 years • Median of 3 prior treatments (Furman, Sharman et al. 2013)	81	NA	NR	NA	AEs ≥20% (any grade): pyrexia (29%), fatigue (24%), nausea (24%), chills (22%), infusion-related reactions (16%), cough (15%)	Oral plus injection; PI3Kδ inhibitor plus anti-CD20 monoclonal antibody	Investigational	
R-CHOP								
NA	NA	NA	NA	NA	NA	Oral + injection MOA varies	FDA-approved for the treatment of diffuse large B-cell, CD20-positive NHL	NCCN 2A

Rituximab							
NA	NA	NA	NA	NA	NA	Injection Anti-CD20 monoclonal antibody	FDA-approved as FCR for previously untreated and previously treated CD20+ CLL

† ESMO III-B: Prospective cohort studies, strong or moderate evidence for efficacy but with a limited clinical benefit, generally recommended; NCCN Category 2A: Based upon lower-level evidence, there is uniform NCCN consensus that the intervention is appropriate

[a] The CHMP granted alemtuzumab positive opinion in multiple sclerosis in June 2013 (EMA 2013).

AE = adverse event; CLL = chronic lymphocytic leukemia; CR = complete response; DOR = duration of response; ESMO = European Society for Medical Oncology; FC = fludarabine, cyclophosphamide; FCR = fludarabine, cyclophosphamide, rituximab; NA = not available; NCCN = National Comprehensive Cancer Network; NHL = non-Hodgkin lymphoma; NR = not reported; OS = overall survival; ORR = overall response rate; PFS = progression-free survival; R/R = relapsed or refractory; SIE/SIES/GITMO = Società Italiana di Ematologia/Società Italiana di Ematologia Sperimentale/Gruppo Italiano Trapianto di Midollo Osseo

5 INTRODUCTION TO ALPHABETINIB

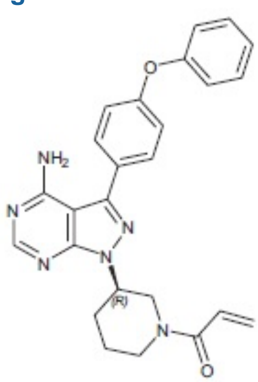
SUMMARY	Alphabetinib is an irreversible inhibitor of Bruton's tyrosine kinase (BTK) that blocks downstream B-cell receptor activation (Akinleye, Chen et al. 2013; Burger and Buggy 2013) In vitro and in vivo studies confirm the specific activity of alphabetinib against BTK-restricted targets (Akinleye, Chen et al. 2013; Burger and Buggy 2013) Alphabetinib has demonstrated clinical activity in phase 1 and 2 clinical trials, with durable remissions against a variety of B-cell malignancies, including MCL, CLL, follicular lymphoma, and diffuse large B-cell lymphoma (Advani, Buggy et al. 2013; Byrd, Furman et al. 2013; Wang, Rule et al. 2013) Alphabetinib is indicated for the treatment of patients with CLL who have received at least one prior therapy Alphabetinib is conveniently administered as a once-daily oral administration taken at the same time each day

5.1 Product overview

5.1.1 Description

The chemical name for alphabetinib (PCI-32765) is 1-[(3R)-3-[4-amino-3-(4-phenoxyphenyl)-1H-pyrazolo[3,4-d]pyrimidin-1-yl]-1-piperidinyl]-2-propen-1-one, and its chemical structure is shown in **Error! Reference source not found.**.

Figure 6: Chemical structure of alphabetinib



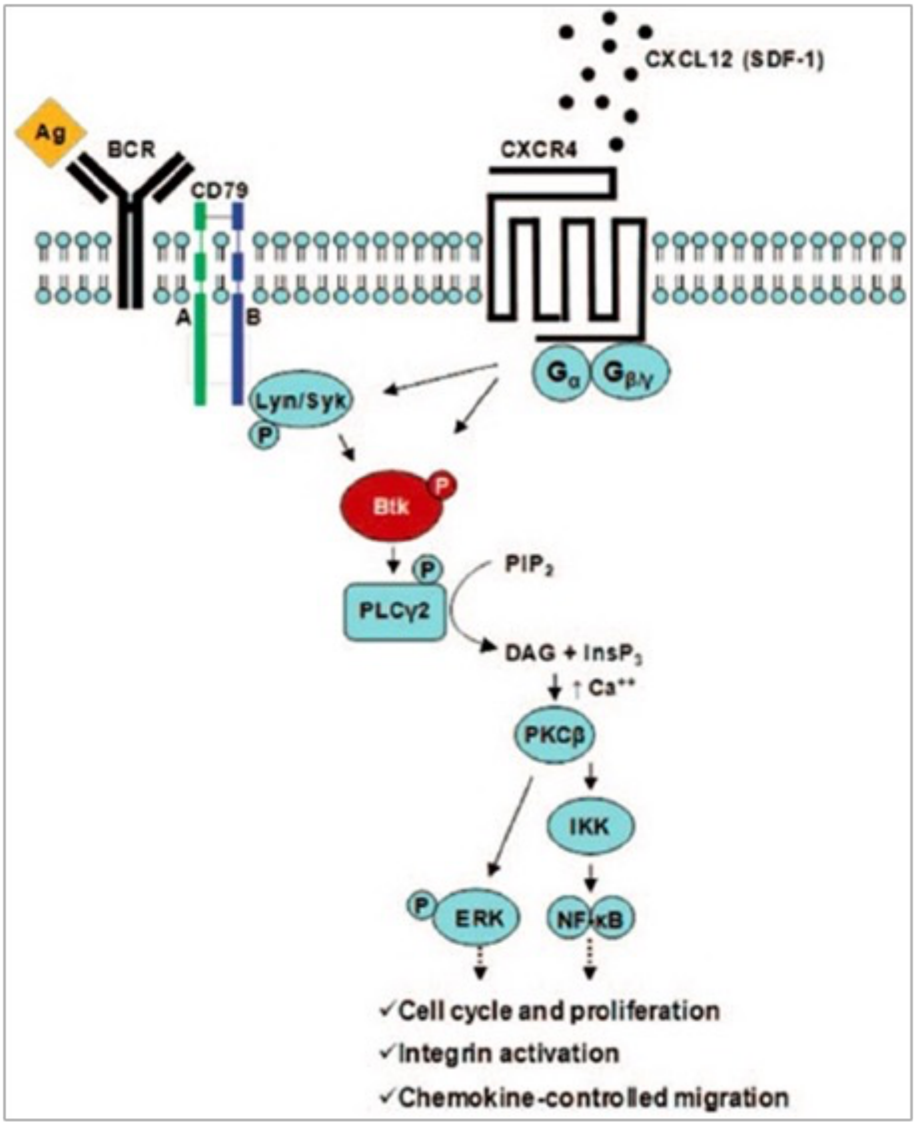
SOURCE: (Pharmacyclics Inc [US PI] 2013)

5.1.2 Mechanism of action

- Alphabetinib is a first-in-class, covalent inhibitor of BTK, a critical signaling kinase in the B-cell antigen receptor (BCR) pathway for tumor cell survival and proliferation (Akinleye, Chen et al. 2013; Burger and Buggy 2013).
- Alphabetinib forms a stable covalent bond with a cysteine residue (Cys-481) in the BTK active site, leading to sustained inhibition of BTK enzymatic activity.

- BTK is an important signaling molecule of the BCR and cytokine receptor pathways (**Error! Reference source not found.**).
- The BCR pathway is implicated in the pathogenesis of several B-cell malignancies, including MCL, diffuse large B-cell lymphoma, follicular lymphoma, and CLL (Furman, Sharman et al. 2013).
- BTK's pivotal role in signaling through the B-cell surface receptors results in activation of pathways necessary for B-cell trafficking, chemotaxis, and adhesion (Pharmacyclics Inc [US PI] 2013).

Figure 7: BTK signaling pathways



SOURCE: (Burger and Buggy 2013)
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- BTK inhibition blocks BCR signaling, induces apoptosis, and blocks migration and adherence (Burger and Buggy 2013; Pharmacyclics Inc [US PI] 2013).

- In preclinical studies, alphabetinib effectively inhibited malignant B-cell proliferation and survival in vivo as well as cell migration and substrate adhesion in vitro. Specifically, alphabetinib inhibits:
  - BCR activation of NF-kB (apoptosis) (Herman, Gordon et al. 2011)
  - BCR activation of integrins (adhesion) (de Rooij, Kuil et al. 2012)
  - chemokine networks (migration/homing) (Ponader, Chen et al. 2012).

5.2 Indications and dosage

Section to be updated upon approval of CLL indication.

5.2.1 Indications

Alphabetinib is indicated for the treatment of patients with CLL who have received at least one prior therapy. .

5.2.2 Dosage

For CLL, the recommended dose is 420 mg (three 140 mg capsules), which should be taken orally once daily, at approximately the same time each day. Alphabetinib should be swallowed whole with water and should not be opened, broken, or chewed.

For all other information about alphabetinib, see the approved product label in your location.

5.3 Clinical evidence for alphabetinib in relapsed or refractory CLL

CALLOUT: PFS as surrogate endpoint

Numerous challenges associated with the application of overall survival (OS) as an endpoint have been addressed in various clinical and regulatory settings. OS is defined as the time from random assignment to the date of death due to any cause (Gutman, Piper et al. 2013). The following disadvantages of OS as an endpoint have been described in the FDA Guidance for Industry, Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics (FDA 2007):

- May involve larger studies
- May be affected by crossover therapy and sequential therapy
- Includes noncancer deaths

The use of outcomes other than OS, including PFS, has attracted growing interest over the past decade. PFS is defined as the time from random assignment in a clinical trial to disease progression or death from any cause (Gutman, Piper et al. 2013). With independent radiological review now frequently required by the FDA and EMA, PFS is considered to be a reproducible endpoint. In several large phase

3 trials, little difference has been found between independent, centrally reviewed PFS and investigator-assessed PFS, supporting the validity of PFS as an endpoint (Escudier 2011).

Beauchemin and colleagues performed a systematic literature review to investigate the relationship between PFS and OS in CLL. The authors included 23 articles among 1,263 potentially relevant studies identified by the literature search. The mean *N* in the studies was 118 patients (minimum, 30; maximum, 724), with a median age=63.0 years, median follow-up=33.7 months, median PFS=14.0 months (SD=12.4), and median OS=35.0 months (SD=31.2). The literature review results showed a significant correlation between median PFS and median OS observed in the second line and subsequent line therapy (Spearman's correlation coefficient = 0.813) ($P \leq 0.001$). The authors' findings strongly support the use of PFS as an effective surrogate for OS in the evaluation of the clinical efficacy of new drugs in CLL (Beauchemin, Johnston et al. 2012).

5.3.1 Phase 1b/2 RESCUE

RESCUE, is a multicenter, phase 1b/2 fixed-dose study of single-agent alphabetinib in treatment-naïve and relapsed or refractory CLL (Byrd, Furman et al. 2013). **Table 14** presents key features of PCYC-1102 design and results. The study duration was 31 months.

Table 14: Key features of PCYC-1102

PCYC-1102
Aim
• To evaluate the efficacy and safety of single-agent alphabetinib in patients with R/R CLL/SLL
Trial design
• Multicenter, open-label, single-agent, phase 1b/2 study of 51 patients ≥18 years of age with R/R CLL/SLL • Two cohorts of patients received 420 mg alphabetinib daily* – Cohort 1 (*n*=27): patients with R/R CLL/SLL – Cohort 4 (*n*=24): patients with high-risk R/R CLL/SLL (subjects with a suboptimal response to at least 2 cycles of chemoimmunotherapy, defined as progression of disease within 24 months of initiation of a regimen containing a nucleoside analogue or bendamustine in combination with a monoclonal antibody or failure to respond to such a regimen) • Alphabetinib administered orally (PO) at 420 mg daily (QD) continuously until disease progression • Treatment with alphabetinib was held for any unmanageable AE that was either potentially study drug-related or Grade 3 or higher in severity; study drug was held for a maximum of 28 consecutive days, and subjects were discontinued from treatment for required delays longer than 28 days • Response assessment was performed by the investigator and followed the IWCLL 2008 guidelines[a]

PCYC-1102
Primary endpoint results

- Primary endpoint: safety assessed according to the frequency and severity of AEs
- Safety outcomes:
  - Predominantly grade 1 or 2 AEs; the most common AEs were diarrhea (59%), URTI (39%), fatigue (33%), and most AEs resolved without the need to suspend treatment
  - Most common of grade 3 or higher were pneumonia (in 4 patients [7.8%]) and hypertension (in 4 patients [7.8%]); infections of grade 3 or higher occurred most frequently early in the course of therapy
  - Patients were able to receive extended treatment with minimal hematologic side effects
  - Only 4% of patients who received 420 mg alphabetinib daily discontinued treatment due to AEs

Secondary endpoint results

- Secondary endpoints: ORR, PFS, pharmacodynamics, PK
- ORRs were 77.8% (Cohort 1: 420 mg R/R CLL) and 79.2% (Cohort 4: 420 mg high-risk R/R CLL)[a]
- In Cohort 1, the CR was 7% with single-agent alphabetinib and the PFS rate was 83% at 24 months (NexGenvion Pharmaceuticals 2013)
- Responses were independent of clinical risk factors present before treatment, including advanced-stage disease and the number of previous therapies
- Pharmacodynamics outcome: median BTK occupancy at 24 hours after drug administration remained >90% in all cohorts and did not fluctuate with plasma drug concentrations
- PK outcome: exposure to alphabetinib increased proportionally from a dose of 420 mg to 840 mg per day

* A total of five cohorts were analyzed in the study; only the results from the two cohorts of patients with R/R CLL who received 420 mg alphabetinib daily are discussed in this GVD; other cohorts included Cohort 2 (Treatment-naïve ≥65 years, 420 mg alphabetinib daily), Cohort 3 (R/R CLL, 840 mg alphabetinib daily), and Cohort 5 (Treatment-naïve ≥65 years, 840 mg alphabetinib daily)

[a] For CLL subjects, treatment-related lymphocytosis was not considered progression and partial response with lymphocytosis was assessed according to clarification of the IWCLL criteria (Hallek, Cheson et al. 2012). For SLL subjects, the International Working Group for non-Hodgkin's Lymphoma 2007 criteria were used (Cheson, Pfistner et al. 2007).

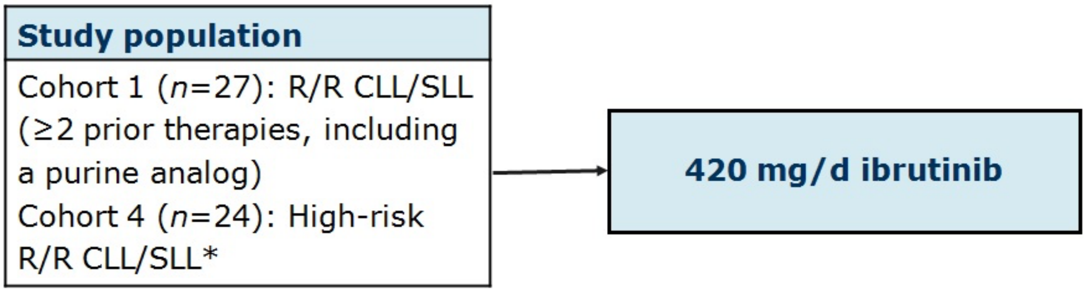
AE = adverse event; BTK = Bruton's tyrosine kinase; CLL = chronic lymphocytic leukemia; ORR = overall response rate; PFS = progression-free survival; PK = pharmacokinetics; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma; URTI = upper respiratory tract infection.

SOURCE: 1102 CSR (Pharmacyclics Inc 2013)

5.3.1.1 Study design

Figure 8 summarizes the PCYC-1102 trial design.

Figure 8: PCYC-1102 study design



* High-risk disease defined as subjects with a suboptimal response to at least 2 cycles of chemoimmunotherapy, defined as progression of disease within 24 months of initiation of a regimen containing a nucleoside analogue or bendamustine in combination with a monoclonal antibody or failure to respond to such a regimen
CLL = chronic lymphocytic leukemia; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: 1102 CSR (Pharmacyclics 2013)

5.3.1.2 Baseline characteristics

Baseline demographic and clinical characteristics of the patients are shown in **Table 15**.

Table 15: Patient baseline demographic and clinical characteristics

Characteristic	Cohort 1 (*n*=27)	Cohort 4 (*n*=24)
Median age, years	64	68
Sex, no. (%)		
Male	20 (74)	17 (71)
Female	7 (26)	7 (29)
Diagnosis, no. (%)		
CLL	26 (96)	22 (92)
SLL	1 (4)	1 (8)
Months since initial diagnosis, median	106.2	81.9
Rai classification at baseline, no. (%)		
Low risk	1 (4)	0 (0)
Intermediate risk	13 (48)	7 (29)
High risk	13 (48)	16 (67)
Missing/unknown	0 (0)	1 (4)
Number of prior induction/salvage therapy, median	4	4
Previous therapy, no. (%)		
Nucleoside analogue	27 (100)	20 (83)
Rituximab	26 (96)	24 (100)
Alkylator	23 (85)	21 (88)
Alemtuzumab	3 (11)	8 (33)
Bendamustine	8 (30)	12 (50)
Ofatumumab	4 (15)	6(25)
Idelalisib	0 (0)	3 (13)
Bulky nodes, no. (%)		
≥5 cm diameter	12 (44)	11 (46)
≥10 cm diameter	0 (0)	3 (13)

Characteristic	Cohort 1 (*n*=27)	Cohort 4 (*n*=24)
IGHV, no. (%)		
Mutated	6 (22)	5 (21)
Unmutated	20 (74)	18 (75)
Data missing	1 (4)	1 (4)
Interphase cytogenetic abnormality, no. (%)		
17p13.1 deletion	11 (41)	7 (29)
11q22.3 deletion	9 (33)	7 (29)
ß$_2$-microglobulin level, no. (%)		
>3 mg/L	9 (33)	9 (38)
Data missing	2 (7)	1 (4)

SOURCE: 1102 CSR (Pharmacyclics 2013)

The median number of prior induction/salvage regimens was 3 (range 1 to 11).

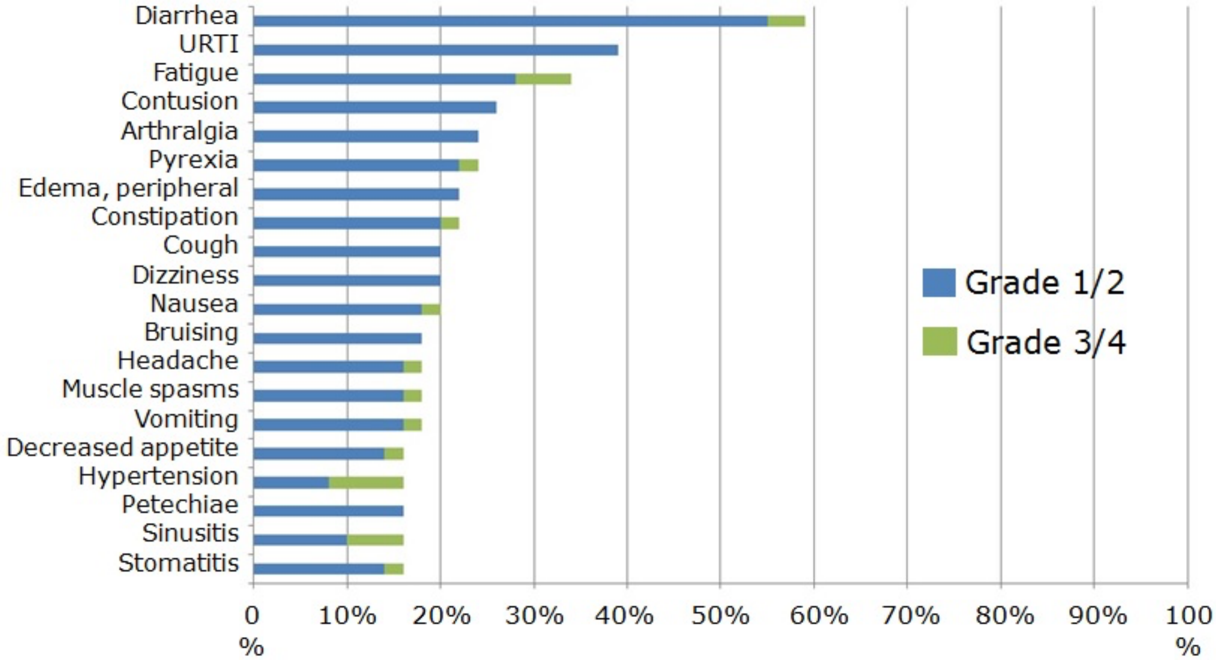
With a median follow-up of 20.9 months (range 0.7 to 26.7), 64% were still receiving treatment, and 36% had discontinued. The reasons for discontinuation included progression (*n*=11; 13%), patient/investigator decision (*n*=13; 15%), and AEs (*n*=7; 8%).

5.3.1.3 Primary endpoint results

Study results for the primary endpoint (safety) showed predominantly grade 1 or 2 AEs. The most common AEs were diarrhea (59%), upper respiratory tract infection (URTI) (39%), and fatigue (33%), and most AEs resolved without the need to suspend treatment. Infections of grade 3 or higher mostly occurred early in the treatment course. The average rate of infection was 7.1 per 100 patient-months during the first 6 months and decreased to 2.6 per 100 patient-months thereafter. Similarly, the exposure-adjusted rate of grade 3 or higher infections lowered by more than half after 6 months of treatment (Byrd, Furman et al. 2013).

Only 4% of patients who received 420 mg alphabetinib daily discontinued treatment due to AEs. AEs of grade 3 or higher were relatively infrequent (**Figure 9**). Patients were able to receive extended treatment with minimal hematologic side effects.

Figure 9: Treatment emergent AEs reported in ≥15% of patients in PCYC-1102



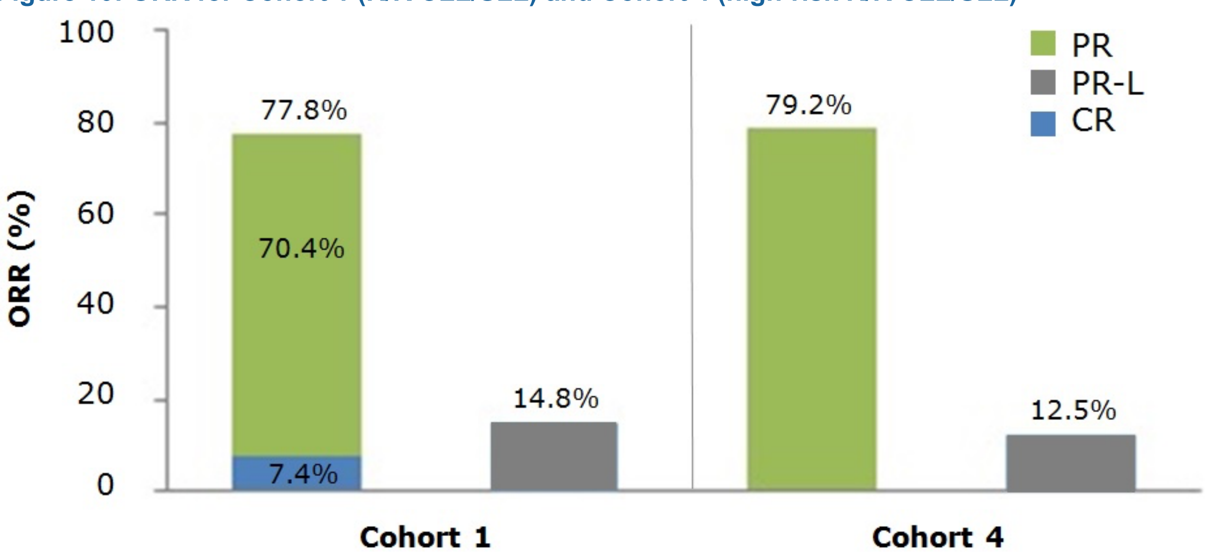
AE = adverse event; URTI = upper respiratory tract infection
SOURCE: 1102 CSR (Pharmacyclics 2013)

Treatment with alphabetinib resulted in the stabilization of immunoglobulin levels (IgG and IgM) and an improvement in IgA with a reduction in the number of patients who needed intravenous immunoglobulin (IVIG) (Byrd, Furman et al. 2013).

5.3.1.4 Secondary endpoint results

A high ORR rate was reported with single-agent alphabetinib (420 mg daily) (**Figure 10**), and the response did not appear to vary according to traditional high-risk prognostic features.

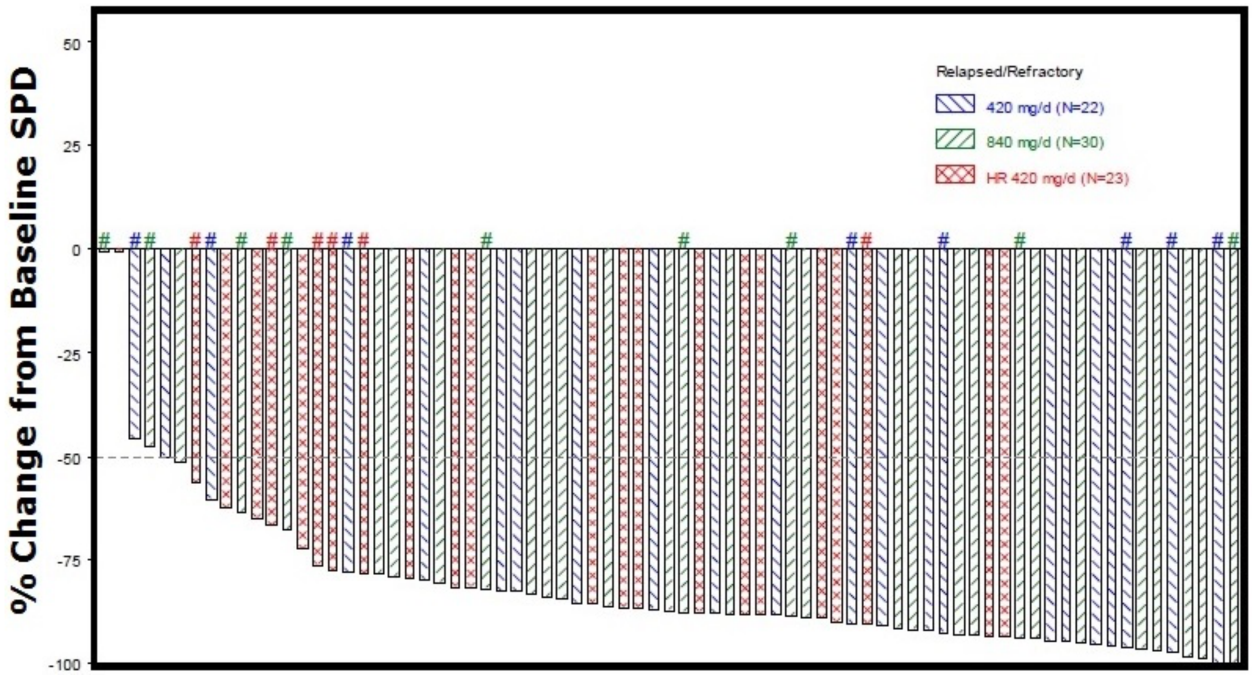
Figure 10: ORR for Cohort 1 (R/R CLL/SLL) and Cohort 4 (high-risk R/R CLL/SLL)



CLL = chronic lymphocytic leukemia; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: 1102 CSR (Pharmacyclics 2013)

Figure 11 shows a waterfall plot of percent change from baseline in the sum of product diameters (SPD) of measurable lymph nodes for patients with relapsed or refractory CLL who had both a baseline and at least 1 post-baseline measurement.

Figure 11: Waterfall plot of percent change in SPD for patients with R/R CLL/SLL



CLL = chronic lymphocytic leukemia; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: 1102 CSR (Pharmacyclics 2013)

The Kaplan-Meier PFS and OS rate estimates are shown in **Table 16**.

Table 16: Kaplan-Meier PFS and OS rate estimates for Cohort 1 (R/R CLL/SLL) and Cohort 4 (high-risk R/R CLL/SLL)

Time point	Cohort 1 (*n*=27)		Cohort 4 (*n*=24)	
	PFS	OS	PFS	OS
6 months	92.1	100.0	91.7	91.7
12 months	92.1	95.8	87.1	91.7
18 months	87.7	91.5	NE	NE/NE
24 months	82.6	91.5	NE	NE/NE

CLL = chronic lymphocytic leukemia; NE = not estimable; PFS = progression-free survival; OS = overall survival; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: 1102 CSR (Pharmacyclics 2013)

The Kaplan-Meier DOR estimates are shown in **Table 17**. The median DOR in Cohorts 1 and 4 has not yet been reached.

Table 17: Kaplan-Meier DOR estimates for Cohort 1 (R/R CLL/SLL) and Cohort 4 (high-risk R/R CLL/SLL)

Time point	Cohort 1 (*n*=21)	Cohort 4 (*n*=19)
Percentage of patients (95% CI) responding		
6 months	100.0 (100.0 to 100.0)	93.8 (63.2 to 99.1)
12 months	94.4 (66.6 to 99.2)	93.8 (63.2 to 99.1)
18 months	88.1 (60.2 to 96.9)	NE
24 months	88.1 (60.2 to 96.9)	NE

CLL = chronic lymphocytic leukemia; DOR = duration of response; NE = not estimable; NR = not reached; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: 1102 CSR (Pharmacyclics 2013)

5.3.2 Open Label Phase 3 RESONATE

PCYC 1112 is a randomized, multicenter, open-label, phase 3 study of alphabetinib versus ofatumumab in relapsed or refractory CLL/SLL. **Table 18** presents key features of PCYC-1112 design and results.

Table 18: Key features of PCYC-1112

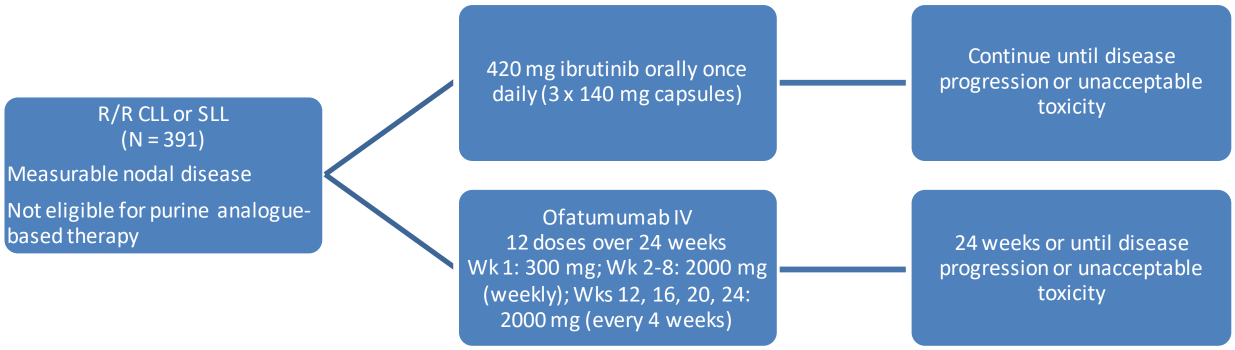
PCYC-1112	
Overview	• Trial designed to evaluate the efficacy and safety of alphabetinib versus ofatumumab in R/R CLL/SLL
Trial design	• Randomized, multicenter, open-label, phase 3 study of alphabetinib versus ofatumumab in R/R CLL/SLL (N = 391) – Alphabetinib 420 mg orally once daily until disease progression or unacceptable toxicity – Ofatumumab IV (Week 1: 300 mg; Weeks 2 through 8: 2,000 mg [once weekly]; Weeks 12, 16, 20, 24: 2,000 mg [every 4 weeks]) for 24 weeks or until disease progression or unacceptable toxicity • Patients in the ofatumumab arm who experience disease progression could receive subsequent therapy with alphabetinib
Primary endpoint results	• Statistically significant improvement in PFS
Key secondary endpoint results	• Statistically significant improvement in OS • Other secondary endpoints: hematological improvements, improvement in disease-related symptoms
Safety findings	• Consistent with previous findings

AE = adverse event; CLL = chronic lymphocytic leukemia; OS =overall survival; PFS = progression-free survival; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: (ClinicalTrials.gov 2013; Pharmacyclics Inc 2014)

5.3.2.1 Study design

Figure 12 summarizes the RESONATE trial design.

Figure 12: RESONATE study design



CLL = chronic lymphocytic leukemia; IV = intravenous; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: (ClinicalTrials.gov 2013; Pharmacyclics Inc 2014)

5.3.2.2 Primary endpoint results

In January 2014, an Independent Data Monitoring Committee (IDMC) unanimously recommended that the Phase III RESONATE trial be stopped early due to significant improvement in primary endpoint of PFS in the alphabetinib arm (Pharmacyclics Inc 2014).

5.3.2.3 Secondary endpoint results

Alphabetinib showed a statistically significant improvement in OS versus ofatumumab (Pharmacyclics Inc 2014).

5.3.2.4 Safety

Early findings from a planned interim analysis of RESONATE indicate that the safety profile of alphabetinib is in line with previous clinical experience (Pharmacyclics Inc 2014).

5.3.2.5 Patient-reported outcomes

5.3.3 Phase 3 placebo-controlled RCT (CLL-3)

CLL-3001 is a randomized, double-blind, placebo-controlled phase 3 study of alphabetinib in combination with bendamustine and rituximab (BR) in relapsed or refractory CLL/SLL. Table 19 presents key features of CLL-3001 design and results.

Table 19: Key features of CLL-3001

CLL-3001	
Overview	• Trial designed to evaluate the efficacy and safety of alphabetinib in combination with BR in R/R CLL/SLL
Trial design	• Randomized, double-blind, placebo-controlled phase 3 study of alphabetinib in combination with bendamustine and rituximab (BR) in R/R CLL/SLL
Primary endpoint results	• Primary endpoint: PFS
Key secondary endpoint results	• Secondary endpoints: safety, ORR, OS, PK
Safety findings	• Most frequently reported AEs

AE = adverse event; BR = bendamustine plus rituximab; CLL = chronic lymphocytic leukemia; ORR = overall response rate; OS = overall survival; PFS = progression-free survival; PK = pharmacokinetics; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: (ClinicalTrials.gov 2013)

5.3.3.1 Study design

5.3.3.2 Primary endpoint results

5.3.3.3 Secondary endpoint results

5.3.3.4 Safety

5.3.3.5 Patient-reported outcomes

5.3.4 PCYC-1117

PCYC-117 is an open-label, single arm, multicenter phase 2 study of alphabetinib in patients with relapsed or refractory CLL/SLL with 17p deletion. **Table 20** presents key features of PCYC-1117 design and results.

Table 20: Key features of PCYC-1117

PCYC-1117	
Overview	• Trial designed to evaluate the efficacy and safety of single-agent alphabetinib in patients with R/R CLL/SLL with 17p deletion
Trial design	• Open-label, single arm, multicenter phase 2 study of alphabetinib in patients with R/R CLL/SLL with 17p deletion
Primary endpoint results	• Primary endpoint: ORR
Key secondary endpoint results	• Secondary endpoints: DOR, PFS, OS, safety
Safety findings	• Most frequently reported AEs

AE = adverse event; CLL = chronic lymphocytic leukemia; DOR = duration of response; ORR = overall response rate; OS = overall survival; PFS = progression-free survival; R/R = relapsed or refractory; SLL = small lymphocytic lymphoma
SOURCE: (ClinicalTrials.gov 2013)

5.3.4.1 Study design

5.3.4.2 Primary endpoint results

5.3.4.3 Secondary endpoint results

5.3.4.4 Safety

5.3.4.5 Patient-reported outcomes

5.4 Place of alphabetinib in relapsed or refractory CLL

Current regimens provide limited efficacy in the treatment of relapsed or refractory CLL, and patients continue to have poor outcomes while also managing the negative impact of poorly tolerated palliative chemotherapy. As such, there is a high unmet medical need for new treatment options in the relapsed or refractory setting.

CALLOUT: Systematic literature review to support cost-effectiveness model for alphabetinib in the treatment of relapsed or refractory CLL

A summary of findings from a systematic literature review to support a cost-effectiveness model for alphabetinib in the treatment of relapsed or refractory CLL is presented in **Table 21**. The review included prospective interventional phase 1, 2, 3, or 4 trials found from searches of PubMed/MEDLINE (January 1, 2001–April 28, 2013) and ASCO/EHA/ASH conference proceedings (2011–2013). Relevant comparators were identified through treatment guideline review and feedback from NexGenvion and their affiliates.

Several data trends and limitations were noted:

- Limited RCT data precludes the use of RCTs in indirect comparison
- The majority of the studies are small single-arm phase I/II–II trials
- Insufficient number of trials per treatment of interest
- Evidence of clinical heterogeneity seen in the accepted trials
- Inconsistent efficacy outcome results within treatments not clearly associated with differences baseline or trial characteristics
- Sparse and heterogeneous progression-free and overall survival data

Table 21: CLL comparators and response rates

Comparator	ORR (%)	CR (%)	PR (%)
Alphabetinib	77.8 to 79.2	0 to 7.4	70.4 to 79.2
Single-agent therapies			
Bendamustine	40 to 93.3	6.7 to 26.7	13.3 to 86.7
Rituximab	25 to 35.9	0	25 to 35.9
Ofatumumab	22.8 to 42.4	0	42.4
Lenalidomide	72.4	13.8	58.6
Combination therapies			
FC	44.6 to 81	2.5 to 25	42.1 to 71.4
FCR	63.9 to 75.4	25.4 to 30.7	35.1 to 44.6
FCM	57.7	7.7	50
FCMR	65.4	15.4	50
Lenalidomide + rituximab	31.8 to 66.1	6.8 to 11.9	25 to 54.9
Ofatumumab + HDMP	81	4.8	76.2
BR	59 to 68.8	9 to 18.8	50
Chlorambucil + prednisone: 3rd line	25	7.5	17.5
Oblimersen + FC	40.8	9.2	31.7

ASCO = American Society of Clinical Oncology; ASH = American Society of Hematology; BR = bendamustine plus rituximab; CLL = chronic lymphocytic leukemia; CR = complete response; EHA = European Hematology Association; FC = fludarabine, cyclophosphamide; FCR = fludarabine, cyclophosphamide, rituximab; FCM = fludarabine, cyclophosphamide, mitoxantrone; FCMR = fludarabine, cyclophosphamide, mitoxantrone, rituximab; ORR = overall response rate; OS = overall survival; PR = partial response; RCT = randomized controlled trial

In patients with relapsed or refractory CLL treated with single-agent alphabetinib, ORRs of 78% to 79% have been observed in phase 2 with durable remissions greater than 2 years, including those with high-risk disease (e.g., subjects with a suboptimal response to at least 2 cycles of chemoimmunotherapy, defined as progression of disease within 24 months of initiation of a regimen containing a nucleoside analog or bendamustine in combination with a monoclonal antibody or failure to respond to such a regimen).

Clinical benefits were shown in difficult-to-treat populations, including the following high-risk groups:

- Patients with *TP53* mutations ± del(17p), del(11q), or complex karyotypes (i.e., more than three cytogenetic aberrations)
- Patients with a short initial duration of remission (<24-36 months)
- Patients who have no response to therapy or progress on therapy

The high response rates, long median PFS, and durable remissions associated with alphabetinib, in combination with the ease of once-daily, oral therapy and low rates of grade 3/4 AEs, provide a compelling clinical value to patients who otherwise have limited treatment options.

For patients with relapsed or refractory CLL, alphabetinib provides an overall positive efficacy and safety profile via a simple once daily oral therapy, with high response rates, prolonged PFS, and durable remissions. Alphabetinib has low rates of treatment-related AEs that may also limit costs relative to current treatments associated with AE management and the requirement for inpatient treatment.

Patients treated with alphabetinib benefit from avoiding chemotherapy for two primary reasons: 1) prevention of chemotherapy-related AEs; and 2) reduced likelihood of long-term development of second primary malignancies[*] compared with traditional chemotherapy. The latter benefit is particularly important in earlier lines of therapy and in younger patients where long-term risk of second primary malignancies is a key consideration and life expectancy is significantly reduced (Ferrajoli 2010).

Unlike other treatments for relapsed or refractory CLL, alphabetinib has been shown to stabilize or restore the normal immune system over time, leading to lower numbers of infections, fewer patients requiring intravenous immunoglobulin (IVIG) administration, and decreased cytopenias (Byrd, Furman et al. 2013; O'Brien, Furman et al. 2014).

Overall, the efficacy and safety profile of alphabetinib will limit costs relative to current treatments associated with treatment-related AEs, inpatient treatment, and less effective palliative therapies for patients with relapsed or refractory CLL.

In summary, few effective therapies are available for patients with relapsed or refractory CLL. Single-agent alphabetinib offers an improved efficacy and safety profile, resulting in longer remissions and

[*] Other malignancies (5%) have occurred in patients who have been treated with alphabetinib, including skin cancers (4%), and other carcinomas (1%) Pharmacyclics Inc [US PI] (2013). IMBRUVICA™ [package insert]. Sunnyvale, CA.

survival, the resolution of underlying treatment-related morbidity, minimization of treatment-related AEs, and enhanced outcomes for patients with relapsed or refractory CLL.

6 ECONOMIC EVALUATION OF ALPHABETINIB IN RELAPSED OR REFRACTORY CLL

6.1 Summary

Summary of CLL model to be added when available.

6.2 Methods

Section to include: modeling approach and model structure, comparators, treatment pathways, model inputs, and model validation.

6.3 Results

Model results for the base case analyses and sensitivity analyses to be added when available.

7 APPENDICES

7.1 Q&A

To be drafted pending discussion with the review team.

7.2 CLL literature search strategies

Table 22: PubMed search strategies – CLL

1	("chronic lymphocytic leukemia"[MeSH Terms] OR "chronic lymphocytic leukemias"[MeSH Terms]) OR “leukemia, chronic lymphocytic”[MeSH Terms] OR “leukemia, chronic lymphocytic”[MeSH Terms])AND "epidemiology"[All Fields]	1256	392	21317446 21239775 21047487 20555070 20008228 19850738 19454423
2	("chronic lymphocytic leukemia"[MeSH Terms] OR "chronic lymphocytic leukemias"[MeSH Terms]) OR “leukemia, chronic lymphocytic”[MeSH Terms] OR “leukemia, chronic lymphocytic”[MeSH Terms])AND "quality of life"[All Fields]	96	36	22498987 21609654 22242823 22234621 22168274 22143060 21047488 21047487 20194844 19197731 18656259
3	("chronic lymphocytic leukemia"[MeSH Terms] OR "chronic lymphocytic leukemias"[MeSH Terms]) OR “leukemia, chronic lymphocytic”[MeSH Terms] OR “leukemia, chronic lymphocytic”[MeSH Terms])AND ("economics"[MeSH Terms] OR "economics"[All Fields] OR "economic"[All Fields])	57	21	22111902 19016733

* Limits: Published in the last 5 years; Humans; English language

Table 23: Regulatory searches – CLL

Source	Search string or *search methodology*
General sources	
Food and Drug Administration (FDA)	*Hand search through 'Oncologic Drugs Advisory Committee' menu options* *Hand search through 'Drug approvals and databases' menu options* *Hand search through 'Drug safety and availability' menu options* *Hand search through 'Development & Approval Process (Drugs)' menu options* *Hand search through 'Newly added guidance documents' menu options* *Hand search through 'Guidances (drugs)' menu options* *Hand search through 'What's New from the Office of Oncology Drug Products' menu options* *Hand search through 'Guidance, Compliance & Regulatory Information' menu options* *Hand search through 'News & Events' menu options*
National Guideline Clearing House	Chronic lymphocytic leukemia
European Medicines Agency (EMA)	Chronic lymphocytic leukemia *Hand search of 'Scientific Guidelines for Human Medicinal Products - Clinical Efficacy and Safety Guidelines'* *Hand search of 'Regulatory and procedural guidance'*
European Observatory on Health Systems and Policies	Chronic lymphocytic leukemia
European Network for Health Technology Assessment	Chronic lymphocytic leukemia
Organisation for Economic Cooperation and Development (OECD)	Chronic lymphocytic leukemia

Table 24: Payer searches – CLL

Source	Search string or *search methodology*
Evidence-based medicine/HTA sources	
EMBASE	Search of Embase using search terms 'chronic lymphocytic leukemia' AND 'epidemiology'; 'chronic lymphocytic leukemia' AND 'quality of life'; 'chronic lymphocytic leukemia' AND 'economic'
Cochrane Library	Hand search of chronic lymphocytic leukemia Search of Cochrane Colloquium Abstracts using search term 'chronic lymphocytic leukemia' in title
HEED	Search of HEED using search terms 'chronic lymphocytic leukemia' AND 'epidemiology'; 'chronic lymphocytic leukemia' AND 'quality of life'; 'chronic lymphocytic leukemia' AND 'economic'
Turning Research into Practice (TRIP)	Chronic lymphocytic leukemia
Clinical Evidence	Chronic lymphocytic leukemia
International Network of Agencies for HTA	Chronic lymphocytic leukemia
www.crd.york.ac.uk	Chronic lymphocytic leukemia
European payers	
Haute Autorite de Sante (HAS)	Search for: chronic lymphocytic leukemia. Hand search of rest of website
Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen (IQWiG)	Hand search of IQWIG publications list (in English) using search term 'chronic lymphocytic leukemia'
Gemeinsame Bundesausschuss (G-BA)	Chronic lymphocytic leukemia
Agenzia Italiana de Farmaco (AIFA)	Chronic lymphocytic leukemia (in English section of website) Hand search of English menu options
Agencia de Evaluación de Tecnologías Sanitarias (AETS)	Hand search through publications list (in Spanish)
National Institute for Health and Clinical Excellence (NICE)	Chronic lymphocytic leukemia *Use of search tool under 'Evidence Search' menu option* *Hand search of 'NICE GUIDANCE*
NHS health technology assessment programme (NCCHTA)	Chronic lymphocytic leukemia *Hand search of 'Published research'* *'Research in progress' menu option* *Hand search 'news' menu option and 'publications' menu option* *Hand search 'Investigators zone' menu option*
Scottish Medicines Consortium (SMC)	Chronic lymphocytic leukemia
Scottish Intercollegiate Guidelines Network (SIGN)	*Hand search of 'Guidelines' menu option*
North American payers	
Agency for Healthcare Research and Quality (AHRQ)	*Hand search of 'Clinical information' menu options including 'Evidence-based practice' option* *Use of search tool under 'Search for Guides, Reviews, and*

Source	Search string or *search methodology*
	Reports' menu option - no limits applied *Hand search of 'publications and products' menu options*
Oregon Health and Science University (OHSU) **Drug effectiveness review project**	*Hand search of 'Publications & Links' menu option* *Hand search of 'Drug Effectiveness Review Project' menu options including 'DERP Publications'* *Hand search of rest of website*
Canadian Agency for Drugs and Technologies in Health (CADTH)	Chronic lymphocytic leukemia *Use of search tool: 'Search HTIS Reports'* *Use of search tool: 'Search All HTA Publications'* *Use of search tool: 'Search CDR Drug Database'*
Health Canada	Chronic lymphocytic leukemia
Ontario Health Technology Advisory Committee (OHTAC)	*Hand search of 'Ontario Health Technology Assessment Series' 'OHTAC Recommendations' 'Decision-Making Framework' 'Publications and Reviews' menu options*
Cancer Care Ontario (CCO)	*Hand search of 'Backgrounders, Fact Sheets and Position Statements' menu and 'cancer drugs' options*
NHS health technology assessment programme (NCCHTA)	Chronic lymphocytic leukemia *Hand search of 'Published research'* *'Research in progress' menu option* *Hand search 'news' menu option and 'publications' menu option* *Hand search 'Investigators zone' menu option*
Australian payers	
Pharmaceutical Benefits Advisory Committee (PBAC) **Pharmaceutical Benefits Scheme (PBS)**	Chronic lymphocytic leukemia *Hand search of 'publications' menu on 'Health and Aging website'* *Hand search of 'health products and medicines' menu on 'Health and Aging website'*

Table 25: Patient and physician resources – CLL

Resource	Search string or *search methodology*
International Union against Cancer (UICC)	'Resources' section of website
European Society for Medical Oncology (ESMO)	'Science & Education' section of website 'Guidelines & Practice' section of website
European Cancer Organisation (ECCO)	'Publications' and 'news' sections of website 'Education' and 'public affairs' sections of website
European Organisation for Research and Treatment of Cancer (EORTC)	'EORTC Bibliography Database' section of website 'News' section of website
Cancer Research UK	'Leukaemia' section of website
Leukemia & Lymphoma Society	'Disease Information & Support' section of website 'Researchers and Healthcare Professionals' section of website
Lymphoma Research Foundation	'Learn' section of website 'Research' section of website
Orphanet	'Rare diseases' section of website
American Cancer Society	Chronic lymphocytic leukemia 'Leukemia' section of website
ASCO	'Meeting Library' section of website
ASH	'Meetings' section of website 'Library' section of website
National Cancer Institute (NCI)	'Cancer Topics' section of website 'Cancer Statistics' section of website
National Comprehensive Cancer Network (NCCN)	'NCCN Guidelines' section of website

CLL = chronic lymphocytic leukemia

7.3 Use of prognostic markers in workup for CLL by region

Table 26: Table of means: Testing for del(17p) by region, country, and practice setting

	Total	Region		Country					Practice Setting		
		US	**EMEA**	**UK**	**France**	**Germany**	**Italy**	**Spain**	Private Practice	Community	Academic
Not tested for del(17p) at diagnosis, %	25.39	18.60	28.85	35.37	32.38	34.40	20.77	20.93	22.01	27.43	21.23
Tested at diagnosis & negative for del(17p), %	57.72	63.76	54.66	49.93	47.51	53.45	63.05	60.05	63.20	54.99	61.05
Tested at diagnosis & positive for del(17p), %	16.88	17.64	16.50	14.70	20.11	12.15	16.18	19.02	14.78	17.58	17.71

Figure 13: Germany–Use of prognostic markers in the diagnostic workup for CLL

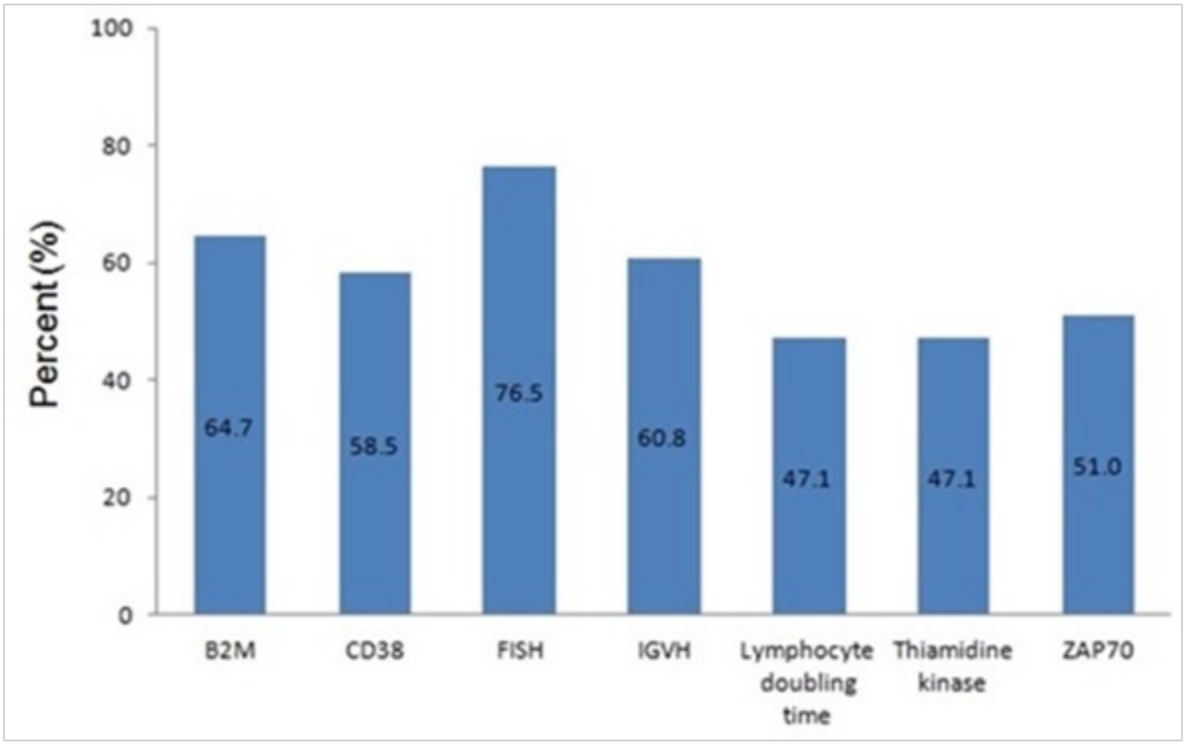


Figure 14: France–Use of prognostic markers in the diagnostic workup for CLL

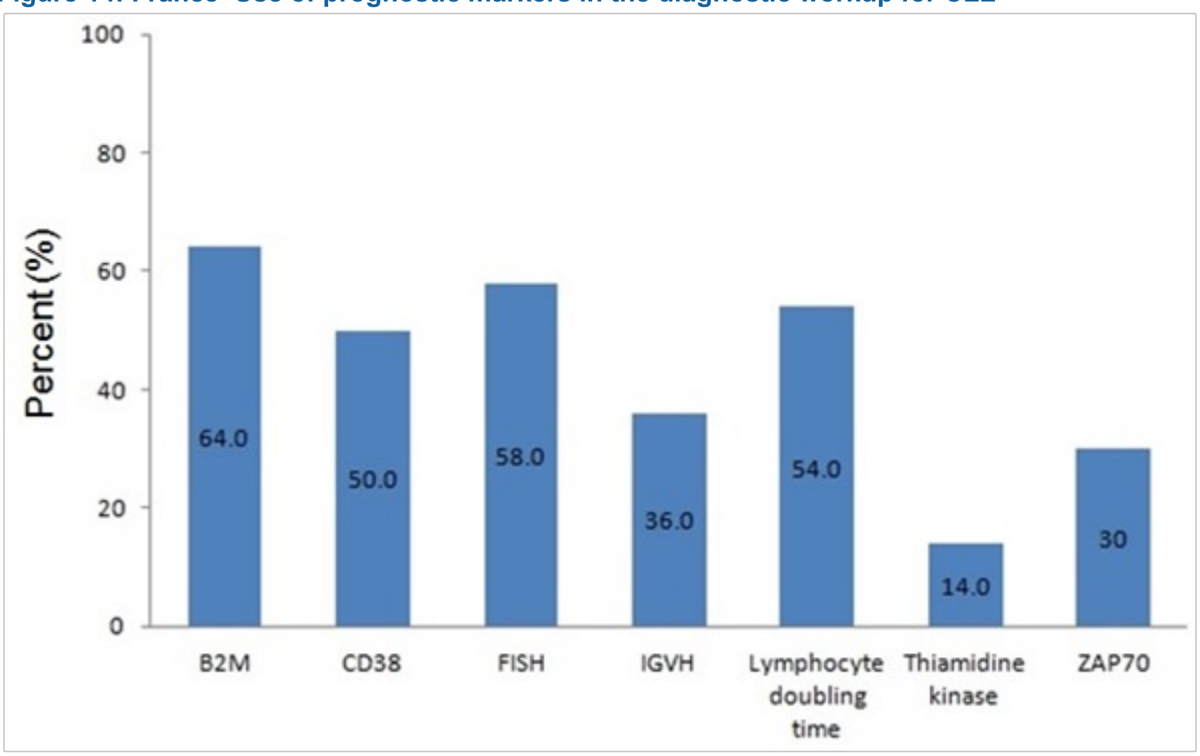


Figure 15: Italy–Use of prognostic markers in the diagnostic workup for CLL

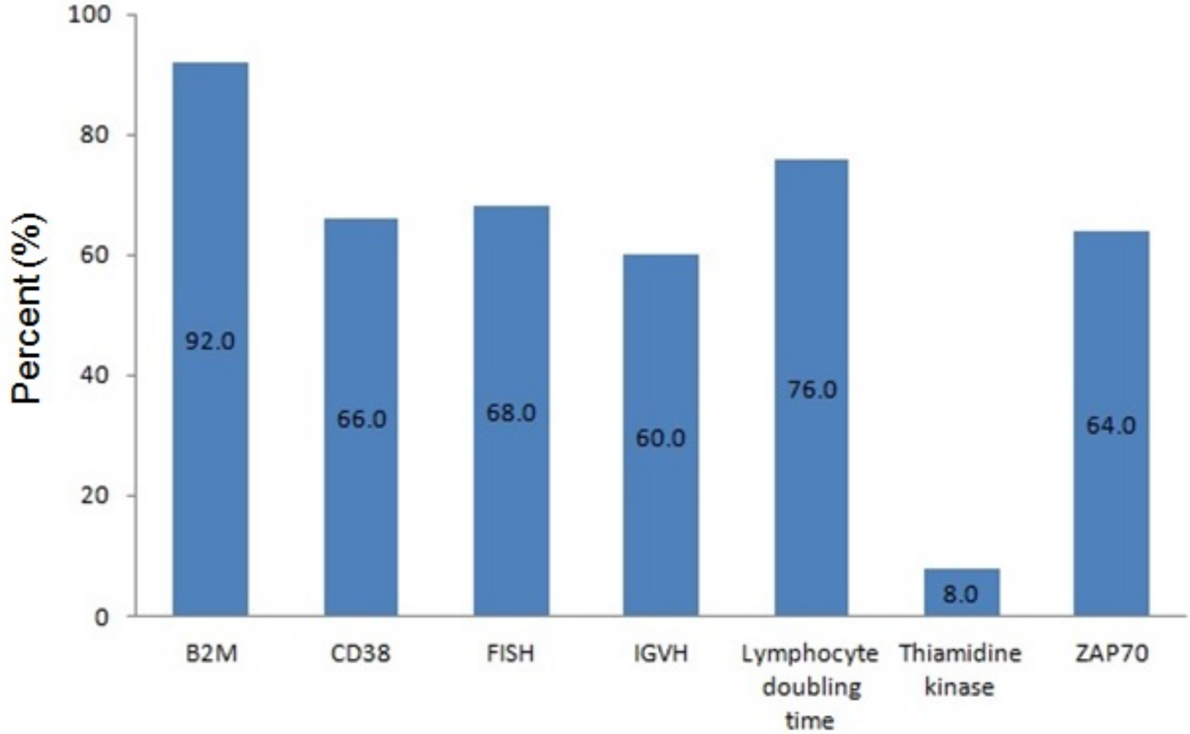


Figure 16: Spain–Use of prognostic markers in the diagnostic workup for CLL

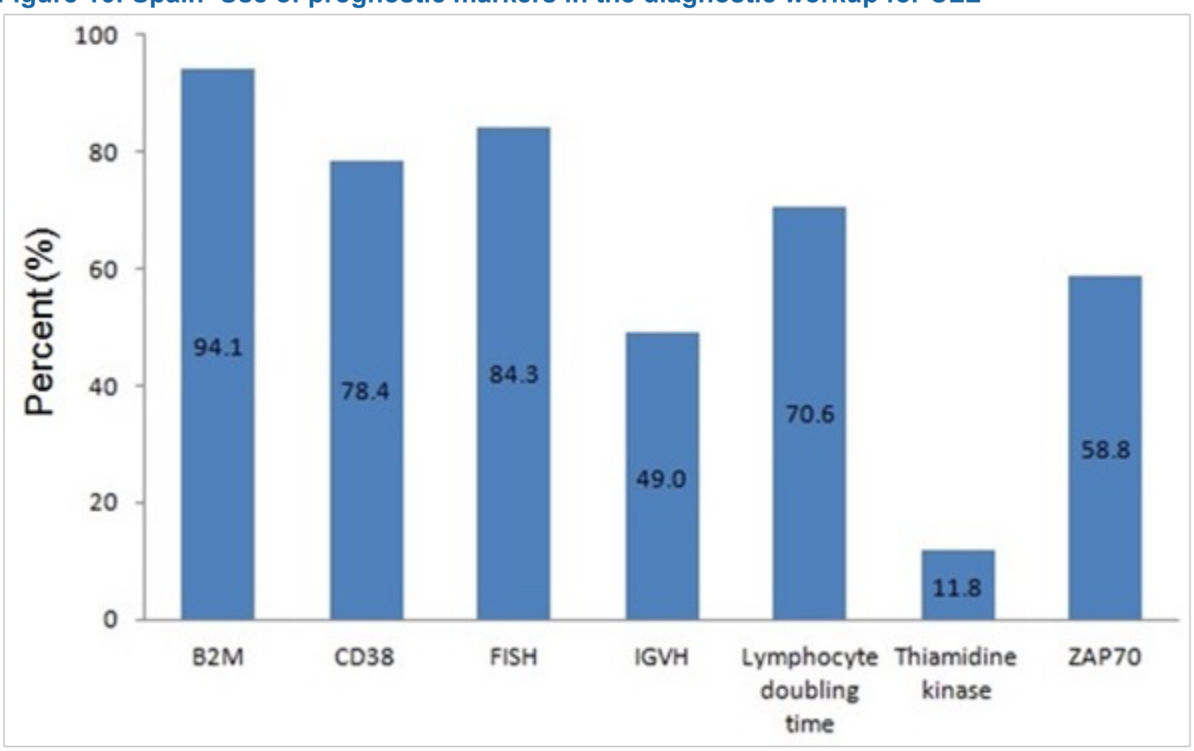


Figure 17: UK–Use of prognostic markers in the diagnostic workup for CLL

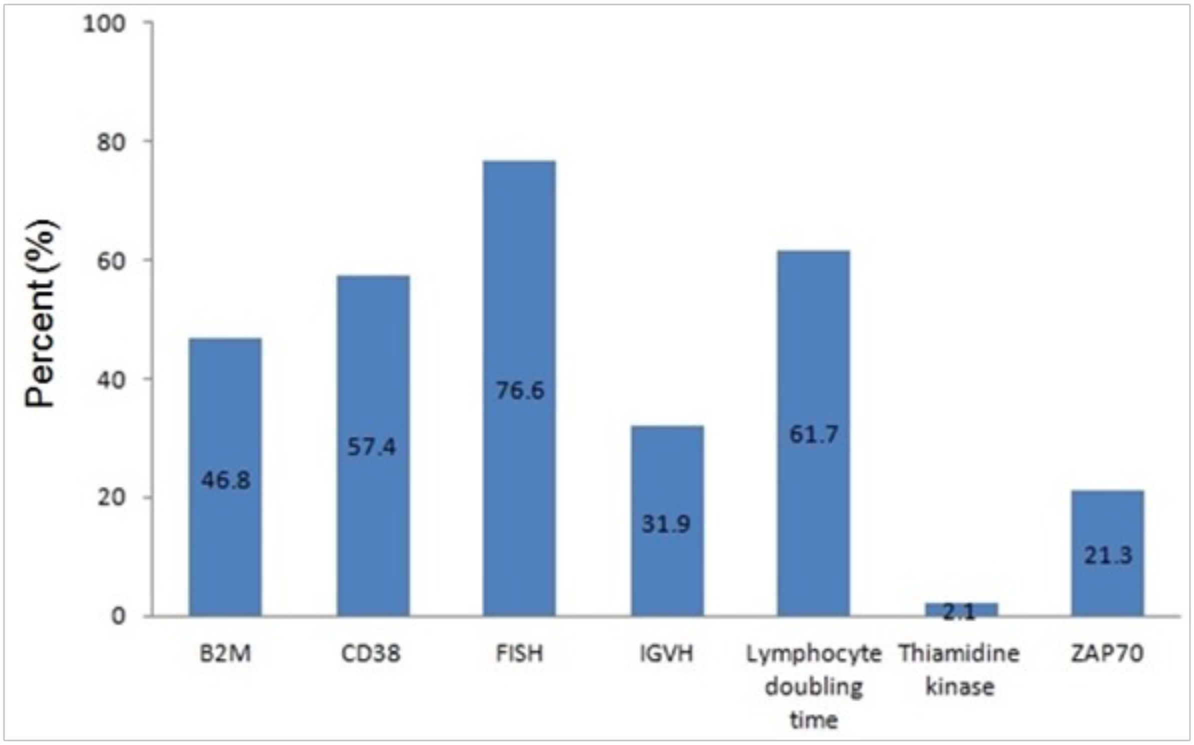


Figure 18: EMEA–Use of prognostic markers in the diagnostic workup for CLL

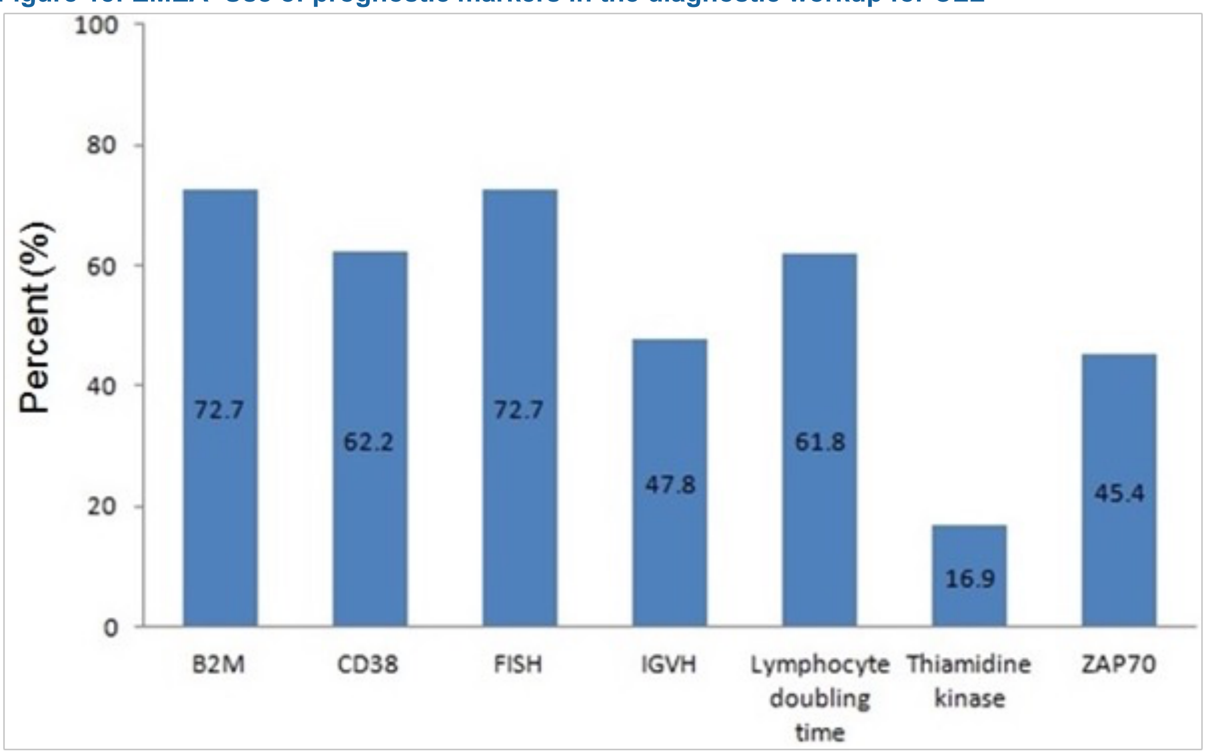
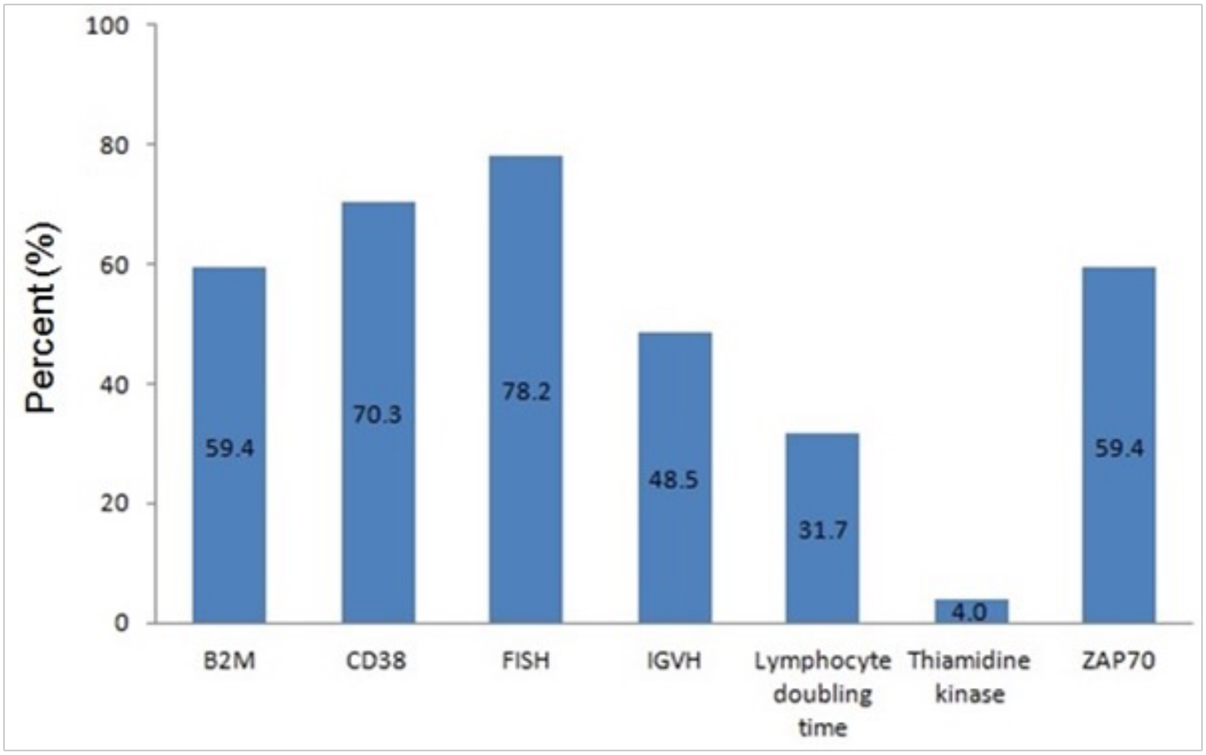


Figure 19: US–Use of prognostic markers in the diagnostic workup for CLL



B2M = ß2-microglobulin; CLL = chronic lymphocytic leukemia; FISH = fluorescence in situ hybridization; IGVH = immunoglobulin variable region heavy chain
SOURCE: (NexGenvion Pharmaceuticals 2013)

7.4 Guideline comparison in relapsed or refractory CLL

Table 27: CLL guideline comparison in relapsed or refractory CLL

ESMO	NCCN	SIE/SIES/GITMO
First-line therapy may be repeated if relapse or progression occurs at ≥12-24 months after monotherapy, or 24-36 months after CIT. If relapse occurs <12-24 months after monotherapy, 24-36 months after CIT, or if the disease does not respond to first-line therapy: • Physically fit patients: salvage regimen, e.g., alemtuzumab[a], followed by allo-SCT • Patients R/R to first-line therapy with alkylating agent: FCR • Physically non-fit patients without del(17p): bendamustine- or alemtuzumab-containing regimen; in subsequent relapses, an attempt with high-dose ofatumumab or rituximab with high-dose steroids can also be made • Patients with del(17p): alemtuzumab-containing regimen	For CLL without del(11q) or del(17p): • In patients with long response to first-line therapy, re-treat as in first line therapy until short response • In patients with short response age ≥70 years, alphabetinib; CIT (reduced-dose FCR, reduced-dose PCR; bendamustine±rituximab; chlorambucil±rituximab; or HDMP+rituximab); ofatumumab; lenalidomide±rituximab; alemtuzumab±rituximab; or dose-dense rituximab • In patients with short response age <70 years or older patients without significant comorbidities, alphabetinib; CIT (FCR, PCR, bendamustine±rituximab; fludarabine+alemtuzumab; R-CHOP; or OFAR); ofatumumab; lenalidomide±rituximab; alemtuzumab±rituximab; or HDMP+rituximab	In patients with no del(17p) and/or p53 mutations, and relapsed after 24 months, the same front-line therapy including rituximab can be considered. In patients with del(17p) and/or p53 mutations, in patients with refractory or relapsed within 24 months from a fludarabine-based treatment, alemtuzumab containing regimens, or experimental treatment approaches within controlled trials should be given. Furthermore, in poor prognosis younger patients with adequate fitness status and no significant comorbidities, a treatment approach including an allo-SCT, either from a sibling or well-matched unrelated donor, should be offered after an appropriate cytoreductive treatment.

[a] Alemtuzumab for the treatment of CLL was withdrawn from the market in 2012 (EMA 2012)

Allo-SCT = allogeneic stem cell transplant; CIT = chemoimmunotherapy; CLL = chronic lymphocytic leukemia; EPOCH-R = etoposide, prednisone, vincristine, cyclophosphamide, doxorubicin, rituximab; ESMO = European Society for Medical Oncology; FCR = fludarabine, cyclophosphamide, and ritixumab; NCCN = National Comprehensive Cancer Network; OFAR = oxaliplatin, fludarabine, cytarabine, rituximab; PCR = pentostatin, cyclophosphamide, rituximab; R-CHOP = rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone; R-hyperCVAD = rituximab, hyperfractionated cyclophsophamide, vincristine, doxorubicin, dexamethasone; R/R = relapsed or refractory; SIE/SIES/GITMO = Società Italiana di Ematologia/Società Italiana di Ematologia Sperimentale/Gruppo Italiano Trapianto di Midollo Osseo

SOURCE: (Eichhorst, Dreyling et al. 2011; Mauro, Bandini et al. 2012; Zelenetz, Gordon et al. 2014)

7.5 Novel agents for the treatment of R/R CLL

Table 28: Novel agents in the treatment of relapsed or refractory CLL as monotherapy

	Alphabetinib	Idelalisib	IPI-145	ABT-199	Ofatumumab	CC-292
N	27 / 24 (R/R / high-risk R/R)	54	44	56	FA-ref (*n*=95), incl. del(17p) (*n*=27)	55
Dose, mg	420	50–350	≤25, 75	150–1200	2000	375–1000
Median prior therapies	4	5	75% with ≥3	4	5	3
Median age, years	64 / 68	63	67	67	64	67
Del(17p), %	41 / 29	24	*TP53* and/or del(17p): 44	29	NR	22
Del(17p), *N*	11 / 7	13	*TP53* and/or del(17p): 32	16	NR	12
Median follow up, months	20.9	NR	NR	NR	NR	12.7/10.4/10.6/ 2.9 (750/ 1000/375/ 500 mg)
ORR (IWCLL PR+CR), %	78 / 79	56	53	85	51	NR
PR by IWCLL criteria, %	68	52	48	72	51	31/57/67/38 (750/1000/ 375/500 mg)
CR, %	7 / 0	4	5	13	0	NR
PR with lymphocytosis, %	15 / 13	11	NR	NR	NR	NR
Duration of response, months	NR	18	NR	NR	NR	NR
PFS	83% at 24 months (R/R)	17 months	NR	NR	5.5 months	NR
OS	92% at 24 months (R/R)	Not reached	NR	NR	14.2 months	NR
Filing date	3Q 2013	4Q 2013	NA	NA	NA	NA
Expected launch	2014	2016	2017	2016	2017	NA
Source	(Pharmacyclics Inc 2013)	(Brown, Furman et al. 2013; ClinicalTrials.gov 2013)	(Flinn, Patel et al. 2013)	(Seymour, Davids et al. 2013)	(Wierda, Kipps et al. 2010)	(Brown, Harb et al. 2013)

CLL = chronic lymphocytic leukemia; CR = complete response; IWCLL = International Workshop on Chronic Lymphocytic Leukemia; NA = not available; NR = not reported; OS = overall survival; PFS = progression-free survival; PR = partial response; R/R = relapsed or refractory

7.6 Epidemiology shells

Table shell for completion by affiliates

Measure	Region	Data	Source
Prevalence, overall CLL			
Prevalence, R/R			
Incidence, overall CLL			
Incidence, R/R			
Percentage of patients with R/R CLL			
Median PFS, overall CLL			
Median PFS, R/R			
Average age at progression into R/R CLL			
Quality-of-life data			
Econ burden data			

Published data

Measure	Region	Data	Source
Prevalence, overall CLL	Europe	3.5 in 100,000	Merchionne et al. 2011
Incidence, overall CLL	Europe	4.92 per 100,000/year	Sant et al. 2010
Incidence, overall CLL	EU5	19,800 (in 2011) 21,300 (est 2016)	Heron BCL Landscape Assessment, 2012
Incidence, overall CLL	US	4.3 per 100,000/year	NCI 2013
Male:female incidence	Europe	5.87:4.01	Sant et al. 2010
Median age @ diagnosis	Europe	72 years	Eichhorst et al. 2011
Median age @ diagnosis	US	72 years	NCI 2013
Median age @ death	US	78 years	NCI 2013
Median OS after failure of first-line treatment	Not specified	1 to 3 years	Tsimberidou et al. 2009; Stilgenbauer et al. 2010
Fatigue and QoL in patients with CLL	International	QoL and BFI scores	Shanafelt et al. 2007
Healthcare utilization in patients with CLL	Germany	Medical costs	Blankart et al. 2013

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